

Sensors for Volatile Organic Compounds

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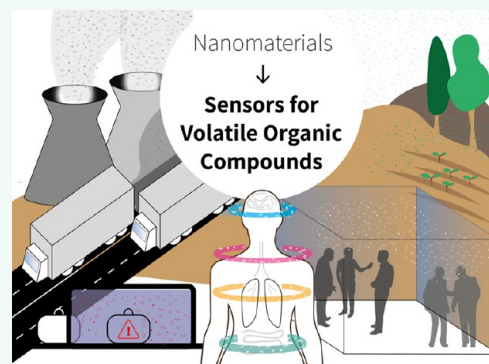
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ABSTRACT: This paper provides an overview of recent developments in the field of volatile organic compound (VOC) sensors, which are finding uses in healthcare, safety, environmental monitoring, food and agriculture, oil industry, and other fields. It starts by briefly explaining the basics of VOC sensing and reviewing the currently available and quickly progressing VOC sensing approaches. It then discusses the main trends in materials' design with special attention to nanostructuring and nanohybridization. Emerging sensing materials and strategies are highlighted and their involvement in the different types of sensing technologies is discussed, including optical, electrical, and gravimetric sensors. The review also provides detailed discussions about the main limitations of the field and offers potential solutions. The status of the field and suggestions of promising directions for future development are summarized.

KEYWORDS: sensor, nanomaterial, volatile organic compound, artificial intelligence, electronic nose, disease, health, biomarker, environment



Volatile organic compounds (VOCs) are chemicals with relatively high vapor pressure at room temperature and atmospheric pressure so they vaporize readily.¹ Sources of VOC emission are widespread and can be of natural origin or through human activities.² They can affect both indoor and outdoor air quality.^{3,4} VOCs can be used as markers of explosives, insect infestation, fruit ripening, and more.^{5–8} Notably, specific VOCs in body fluids and waste (e.g., blood, breath, sweat, urine, feces, etc.) can be used as indicators of diseases and health conditions.^{9–17} Therefore, detection and monitoring of VOCs is expected to find useful applications in multiple areas including health management, environmental monitoring, public safety, agriculture, and food production (Figure 1).

At present, VOC sensing is challenging, mainly because most VOCs are nonreactive and appear in low concentrations, often close to or below the detection limits of analytical or sensing devices. Large laboratory-based or even compact mass spectrometry (MS) is one of the most powerful and widely used characterization tools in scientific and industrial research. It offers comprehensive analysis, quantification, and identification of a wide range of compounds and is well suited to the discovery of VOCs.^{18,19} Currently available MS tools, often coupled with either liquid or gas chromatography (GC), can detect tens of thousands of known and unknown chemical compounds.²⁰ Nevertheless, MS requires relatively bulky, expensive, power- and time-consuming, equipment, and trained personnel.⁵ To address these limitations and to satisfy the requirements for VOC sensing, many other types of sensor

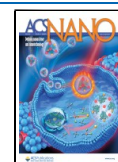
have been made commercially available or established in research laboratories.⁴ However, almost all of them lack the sensitivity, selectivity, and reliability required for the unpredictable VOC combinations found in real samples. The ideal VOC sensor will be highly informative and characterized by high responsivity, good selectivity, fast response/recovery, great stability/reproducibility, ability to work at room temperature, low cost, and ease of fabrication for practical applications.^{21–26} Huge efforts have been dedicated to the pursuit after such ideal sensors in recent decades. This trend has been fueled by progress in nanotechnology and nanomaterials facilitating the design and preparation of advanced sensing materials and devices.^{10,11,15,21,24,25,27–33}

The current paper reviews the field of VOC sensors, focusing on the main research topics and directions. It starts by introducing the various transduction mechanisms, including electrical, optical, and gravimetric. The advantages and disadvantages of each are presented and recent trends that could hold promise for VOC sensing are described. The review then explores the beneficial marriage of VOC sensors with nanomaterials and nanotechnology, which has led to improve-

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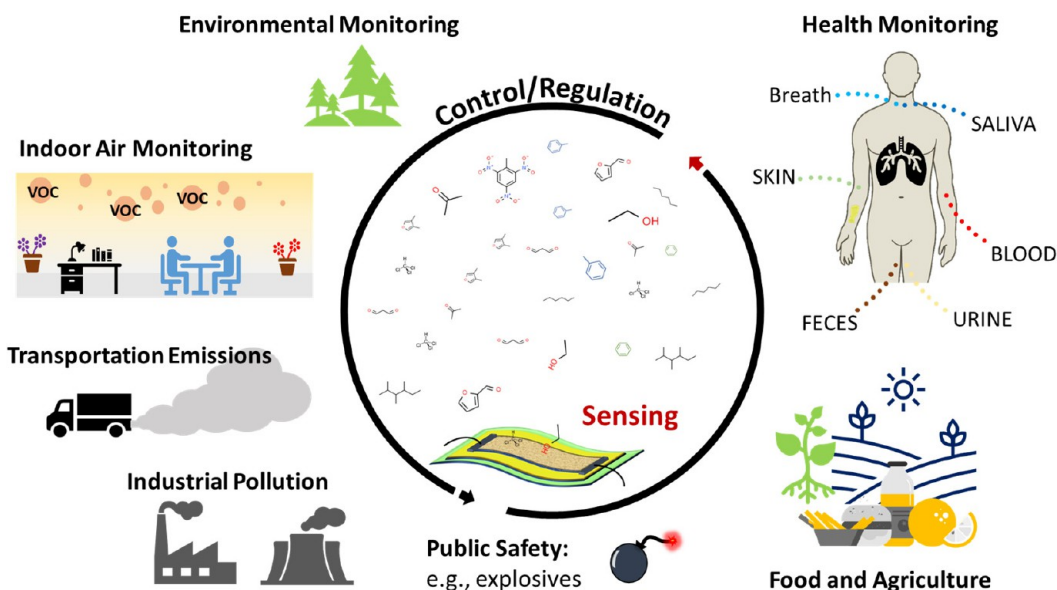


Figure 1. VOC closed loop as part of the IoT network, where VOC emissions are continuously monitored through smart sensing devices. The information obtained from VOC emission and sensing can be used for further control and regulation (e.g., decrease in industrial VOC generation) or making decisions (e.g., in healthcare, existence of a disease and need for treatment). These decisions could be enhanced by artificial intelligence.

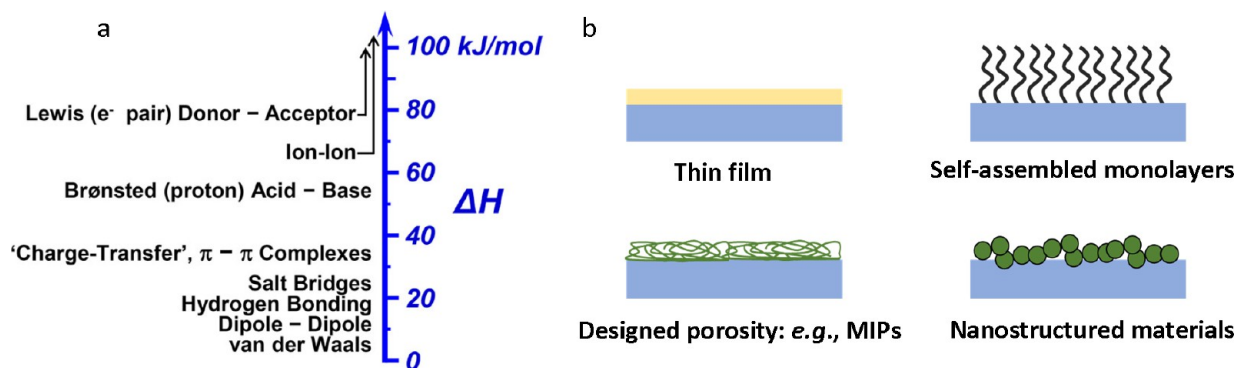


Figure 2. (a) Spectrum of intermolecular interactions that can be used to probe VOC signals presented as absolute values of ΔH . Reprinted with permission from ref 55. Copyright 2013 Royal Society of Chemistry. (b) Examples of receptors that have been widely used to enhance the selectivity and sensitivity of VOC sensing in optical, electrical, and gravimetric devices. Top left: thin film coatings have been widely used with sensing devices such as QCMs, SAWs, and optic fibers. Top right: self-assembled monolayers have been widely used in combinations with inorganic sensing materials such as Au. Bottom left: coatings with designed porosity. Bottom right: nanostructured materials used as receptors to capture and interact with VOCs.

ments in material design and fabrication and sensing performance. At the end of the article, selected topics are described such as artificial nose technologies, the role of VOC sensors in advancing the “internet of things” (IoT) paradigm, and the emerging trend of highly informative sensors.

1. VOC RECOGNITION

1.1. Molecular Interactions of VOCs. Fundamentally, chemical sensing is based on molecular recognition, which is a consequence of interactions between molecules.^{34–36} Intermolecular interactions have a wide range of strengths, from weak van der Waals forces to strong covalent and ionic bonds (Figure 2a). Between those two extremes lie covalent or coordination bond formation, Lewis acid–base interactions, hydrogen or halogen bonding, charge-transfer and π – π stacking complex formation, dipolar or multipolar interactions, and other nonspecific forces.³⁷ VOC sensors that rely on stronger bonds involve sensor–analyte interactions that allow

for more specific detection of analytes, improving the discriminatory power of general sensors.³⁷ In other words, strong VOC–sensor interactions can eventually lead to greater sensitivity and selectivity. However, they can compromise the reversibility of the sensor response.

1.2. Receptors for VOCs. Receptors are an important part of chemical sensors since they are responsible for recognizing the target analytes.³⁸ They are widely used in natural systems for chemical sensing and biosensing. Classically, the word “receptors” denoted biological components such as enzymes, antibodies and nucleic acids. However, more recently, the word has come to denote entities, both natural and synthetic, used to probe chemical interactions.³⁸ The preparation and integration of receptors into VOC sensors have been facilitated by progress in nanotechnology and nanomaterials, which have greatly enhanced the analytical performance of sensors.³⁹ In this section, we will briefly discuss different categories of receptor that have proved useful in VOCs sensing schemes.

Throughout our work, we have noticed that different words have been used to denote receptors for VOC sensing such as selectors,⁴⁰ sensitizers,⁵ and indicators.⁴¹ All these terms describe entities added to a chemical sensing system to impart specific interaction possibilities and enhance sensitivity toward VOCs; hence, they are receptors. In general, receptors for VOC sensors can be categorized as synthetic or biological. Synthetic receptors can appear in different forms including thin films (e.g., polymers and small molecules),^{42,43} nanostructures (e.g., nanotubes and nanoparticles),^{44–46} self-assembled monolayers,^{47,48} and porous matrices (e.g., molecularly imprinted polymers, MIPs) (Figure 2b).^{49,50} They provide selectivity at two different levels: (i) chemical selectivity, based on intermolecular interactions that are mainly defined by the chemical structures of the sensing materials and analytes; and (ii) structural selectivity, which is crucial for recognizing a VOC via the 3D structure of the sensing layer. Self-assembled monolayers have mostly been used with inorganic sensing materials such as metal nanoparticles and silicon nanowires,^{51,52} which show interesting tuning capabilities in the sensing performance. Silane- and thiol-based molecules have received much attention in this area since they are widely available as commercial products and can easily be connected to inorganic surfaces.^{53,54}

While many sensing devices require receptors (e.g., inert/passive quartz-crystal microbalances (QCMs) and optical fibers), some rely on sensing materials with inherent recognition capabilities that can serve as both transducers and receptors. For example, conductive polymers in chemiresistors and FET devices and dielectric polymers in chemicapacitors are the main active components in the transduction mechanism and include organic moieties capable of interacting with VOCs. The same behavior can be found in fluorometric and colorimetric dyes (e.g., metal porphyrins) that offer a plethora of possibilities for interactions with VOCs and centers for optical transduction.

2. SENSING STRATEGIES: TRANSDUCTION MECHANISMS AND SENSOR CLASSIFICATION

In the current section we review the electrical, optical, and gravimetric VOC sensors that continue to engage scientific and technological interest and create challenges. This area includes, but is not limited to, chemiresistors, field-effect transistors (FETs), electrochemical sensors (ESs), QCMs, surface acoustic waves (SAWs), and colorimetric and fluorometric devices.^{10,24} Other sensing methods such as nondispersive infrared sensors (NDIRs) sensors, photoionization detectors (PIDs), and calorimetric sensors will not be discussed in this review.

2.1. Electrical VOC Sensors. Sensors that transduce through electrical signals have been adopted widely owing to their straightforward readout and connection with existing electronic systems. Electrical signals in VOC sensors can be generated in different ways. Therefore, many subtypes of electrically controlled transduction methods exist in combination with a wide range of sensing materials (Figure 3).⁴ Established electrical sensors, including commercially available ones, are not covered in the current review. They are discussed in other articles.⁴

2.1.1. Chemiresistors. Since the 1960s,^{57,58} chemiresistors consisting of sensing material bridging a gap between two electrodes have become promising candidates for VOC sensing. They have numerous advantages,^{21–24,27,59,60} includ-

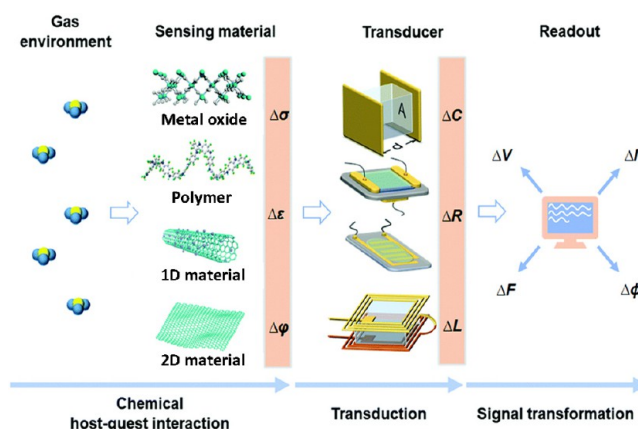


Figure 3. Schematic illustration of the electrical gas sensing process. Reprinted with permission from ref 56. Copyright 2020 the Royal Society of Chemistry.

ing ease of fabrication, variety of suitable sensitive materials, and simple sensing data, which have ensured successful commercialization in certain applications.⁴ They rely on sensing material that has inherent (baseline) resistance. Upon exposure to individual or mixtures of VOCs, the baseline resistance is modulated via a combination of sensing mechanisms. For example, in monolayer-capped nanoparticle-based chemiresistors, the interaction of VOCs with the organic monolayer modulates the distances between nanoparticles.⁶¹ This changes the resistance in proportion to the amount of VOC and/or the mixture composition, thus enabling the VOC to be detected and quantified. Whereas, in conductive polymer-based chemiresistors, swelling, doping, and protonation mechanisms could explain the changes in resistance upon exposure to VOC.⁵⁶

So far, a wide range of materials have been incorporated in chemiresistors and used for VOC sensing including metal oxides,^{62–66} metal nanoparticles,^{48,67} conductive polymers,^{68–70} inorganic semiconductors (e.g., phosphorene and MXenes),⁷¹ carbon materials,^{28,72} hybrid material and composites,^{24,27} and ionic conductors.⁷³ Nanomaterials in particular have seen most use and still dominate this research area owing to their large surface areas and superb sensitivity.^{24,27,74}

2.1.2. Electrochemical Sensors. Electrochemical sensors use redox reactions to reduce or oxidize VOC molecules following their diffusion to the working electrode interface in an electrochemical cell containing a solid, liquid, or gaseous electrolyte or ionic conductor.⁷⁵ The sensing mechanism depends on the electrochemical reaction at the electrodes, which leads to charge transfer and current flow correlating with VOC concentration.^{76,77} Electrochemical sensors can operate either as amperometry-based, where the redox current is measured, or potentiometry-based, where the difference in potentials is measured.⁷⁸ Cyclic voltammetry can also be performed with electrochemical sensors. Here, the distinct oxidation or reduction signaling peaks obtained for each VOC provide a highly selective sensing approach.⁷⁹

Electrochemical sensors have proven selective, accurate, and reliable for sensing VOCs in various environments.²⁷ There are many examples of such sensors in the literature and even on the market,⁴ and they have been used for food,^{80,81} agriculture,⁸² air pollution,⁸³ safety and security,²⁹ and disease detection.⁸⁴ The ability of these sensors to work both in

gaseous and liquid media increases the spectrum of their applications to more intriguing areas, such as wearable devices for monitoring health biomarkers in sweat.⁸⁵

2.1.3. Field-Effect Transistors. A field-effect transistor (FET) is a device that controls the flow of electrical carriers between two electrodes (source and drain) by regulating the conductivity of a semiconductive channel.^{86,87} When a FET bases sensor is exposed to VOCs, individual or mixed, the device's electrical characteristics (e.g., threshold voltage, subthreshold swing, mobility, etc.) change, allowing VOC detection and quantification.⁸⁸ The result is a multiparametric/multivariable sensing performance that detects a higher and wider range of inherently different signals, making it more usable for VOC recognition than that obtained using a chemiresistor.^{9,51,89–91}

Several types of materials have been used for developing FET-based VOC sensors. These include carbon nanotubes (CNTs),⁹² silicon thin films or nanowires,^{93,94} and polymers.^{88,95} Among these, special attention has been given to silicon-based FET VOC sensors, mainly because of their possible direct integration with existing electronic platforms. For example, Javey and co-workers⁹⁴ reported a chemical-sensitive and highly miniaturized FET platform based on 3.5 nm-thick silicon channel transistors. Using industry-compatible processing techniques, the conventional electrically active gate stack is replaced by an ultrathin chemical-sensitive layer that is electrically nonconducting and is coupled to the 3.5 nm silicon channel. Although to date these have been used to target toxic gases such as H₂S, H₂, and NO₂ rather than VOCs, their technology offers significant advantages. Importantly, the system described can readily be integrated with mobile electronics for distributed sensor networks in environmental pollution mapping and personal air-quality monitors.

Another interesting category of FETs that have gained attention as candidates for VOC sensing is organic FETs (OFETs).^{88,95–99} Organic semiconductors have inherent merits (i.e., variety in molecular design, low cost, lightweight, mechanical flexibility, and solution processability), which, in turn, enable high sensitivity, selectivity, and expeditious responses to be achieved. The underlying operating principle of these devices relies on changes in their electrical characteristics upon exposure to VOCs through one or a combination of the following mechanisms: trapping, doping, altering the molecular arrangement of the organic semiconductor layers, or influencing charge injection/extraction at electrode/organic semiconductor interfaces. Depending on the geometry of the device and the morphology of the organic semiconductor, these interactions could occur on the surface, at the grain boundaries, or at various interfaces in OFETs.^{88,96,98}

2.1.4. Chemicapacitors. Chemicapacitors are devices comprising a dielectric layer sandwiched between two electrodes. Upon exposure to VOCs, the intrinsic capacitance of this structure changes owing to modulation of the dielectric properties and/or thickness.¹⁰⁰ This mode of operation ensures low power consumption and suitability for miniaturization.¹⁰⁰ Although different materials have been used to construct chemicapacitors, nonconducting polymers have received most attention because of their easy processability and film deposition, the wide variety available, and their suitability for chemical modification to tune their affinity toward targeted VOCs. Chemicapacitors have shown selective detection of target molecules at ppm and subppm levels, but

nonspecific absorption and cross-sensitivity to other molecules in the atmosphere constitute big challenges. This is especially true for water, which has a very high dielectric constant.¹⁰⁰ These obstacles could be overcome by cross-reactive sensing arrays consisting of different dielectric films and/or schemes for temperature and humidity calibration via pertinent onboard sensors.

2.2. Gravimetric Devices. One property that can be used for measuring VOCs is their mass. Using gravimetric devices, viz. microelectromechanical systems with mechanical resonance, interactions of VOCs with the sensor's surface through physisorption or chemical reactions are translated to changes in resonant frequency.⁵ Among the many gravimetric devices, those commonly used for VOC detection include SAW resonators, QCMs, bulk acoustic wave resonators (BAWs), film bulk acoustic resonators (FBARs), and cantilever-based sensors.⁵ Generally, while some types of sensors can integrate inherently sensitive/selective materials (e.g., OFETs), gravimetric sensors lack selectivity and sensitivity because they are mostly based on inert inorganic materials such as the zinc oxide and lead zirconate titanate (PZT).⁵ To facilitate the absorption and interaction of VOCs with a gravimetric device, the exposed device's surfaces are coated with receptor elements (often referred to as sensitizers) such as polymers, self-assembled monolayers, MIPs, metal oxide nanostructures and films, CNTs and graphene, and supramolecular structures.⁵ Atmospheric conditions such as temperature, pressure and humidity can influence the performance of a gravimetric device drastically,^{10,101} so they should be tracked and neutralized.

2.2.1. QCM. QCM is a well-established, low-cost, and readily available technology. The resonance frequency of the device is $f = \nu / (2d)$, where ν is the acoustic velocity of quartz and d is its thickness. QCM devices can be used for sensing in both liquid and gas environments, which makes them useful for a wide range of applications.⁵ The simplicity of this system has led to the commercialization of multiple QCM modules. Since QCM VOC sensors rely on mass detection and can be affected by almost any type of nonspecific interaction, they usually offer compromised selectivity.²⁴ As with other gravimetric devices, the ability of QCMs to sense a VOC can be controlled by different coating layers selected from a wide range of MOs,^{102–104} polymers,^{105–109} and other materials.^{110,111} MIPs, which can serve for the selective capture of targeted VOCs,^{112–114} have received particular attention as a coating material for QCM modules.

2.2.2. SAW Sensors. These resonators^{115–117} have been the most popular choice of gravimetric VOC sensing devices because they are simple to produce.¹¹⁸ The operating principle of SAWs relies on a standing wave created at a given resonance frequency between two interdigitated electrodes on the surface of a piezoelectric material. The resonance frequency of these devices is $f = \nu / (4d)$, where d is the width of the interdigitated finger and ν is the acoustic velocity of the chosen piezoelectric material.¹¹⁹ Upon exposure to VOCs, the frequencies of waves traveling along the surface of the SAW device are modulated because of changes in conductivity, stress, mass loading, and a viscoelastic effect.¹⁰¹ Various coating layers for SAW VOC sensors have been used such as MOs,^{120,121} graphene,¹²² polymers,^{123,124} and CNTs.¹²⁵

2.3. Optical Sensors. Optical chemical sensors are a broad class of devices used for detecting chemical interactions via radiation intensity in the infrared, visible, or ultraviolet regions.¹²⁶ Generally, they have three main parts: (i) a light

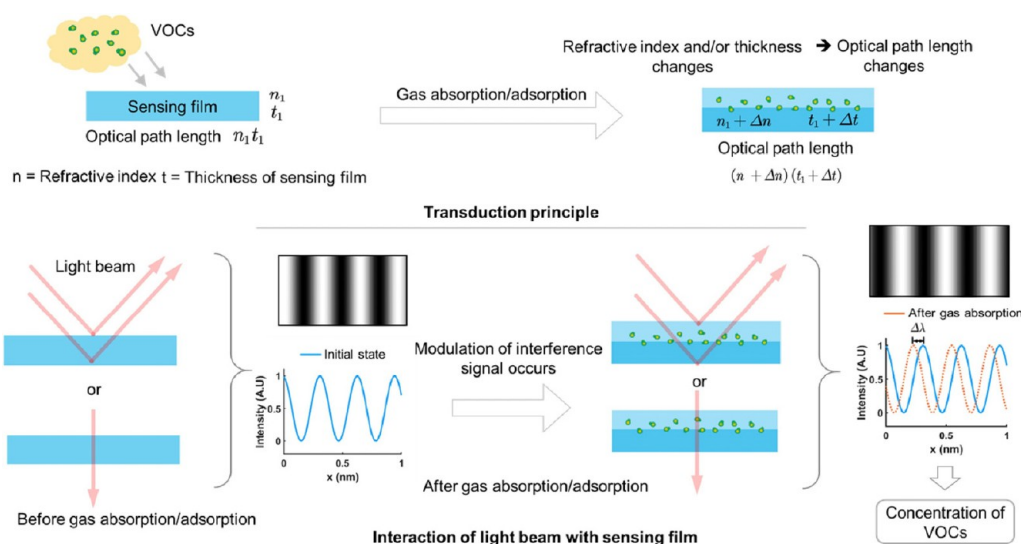


Figure 4. Mechanism of interferometry for gas detection. The sensing film is exposed to VOCs, which change its thickness and refractive index, resulting in a shift of wavelength. Reprinted with permission from ref 152. Copyright 2019 Elsevier B.V.

source; (ii) a substrate or sample cell, where analytes change the amount of radiation absorbed or emitted; and (iii) a detector that detects the wavelength of interest.³⁷ On this basis, optical chemical sensors can be categorized as direct or indicator-mediated sensors. In a direct optical chemical sensor, the analyte is detected via an intrinsic optical property (e.g., absorption). Examples include Raman spectroscopy (SERS), NDIR, and Fourier transform infrared (FTIR) sensors.¹²⁷ In indicator-mediated sensing systems, a change in the optical response of an intermediate indicator, usually a VOC-sensitive dye, is used to monitor the VOC concentration. This approach is most useful when the VOC has no convenient intrinsic optical properties. Another classification of optical sensors is based on the transduction mechanism, such as absorption, scattering, diffraction, reflection, refraction, and luminescence.³⁷ Two of the most common of these VOC sensing approaches are colorimetric and fluorescent. Regardless of the multiple advantages offered by optical sensors, the basic cost of fixed or tunable optical sources (e.g., laser) and the need for an adequate optical path-length and a suitable optical detector with high responsivity and low noise render these systems expensive, demanding on power, and less amenable to miniaturization.¹⁰⁰

2.3.1. Colorimetric Sensors. Colorimetric sensors are optical sensors that change color in response to external stimuli. Any change in the physical or chemical environment can be considered such a stimulus.^{128,129} Colorimetric sensing is straightforward and convenient for use in portable devices, especially with the advent of universal digital color imaging.^{130–132} They can be considered the simplest optical sensors because of the easy readout, which can often be accomplished with the naked eye (qualitatively) or with common smartphone cameras (quantitatively). While many colorimetric sensors use the traditional three-channel visible range (i.e., red, green, and blue, RGB), they can also use more channels with a narrower spectral range for each, combine nonvisible wavelengths from near-IR to UV, or span this full wavelength range using spectrophotometry.³⁷ As of today, many different types of dyes have been used to prepare colorimetric sensors and arrays including Lewis acid–base dyes, Brønsted acidic or basic dyes, and solvatochromic and

vapochromic dyes.⁵⁵ The value of such devices has been demonstrated in optoelectronic nose systems and targeted multiple applications including healthcare, environmental monitoring, and food analysis.⁴¹

2.3.2. Fluorescent Sensors. Fluorescent VOC sensors convert the interaction with VOCs to changes in their fluorescent emission spectra.³⁷ They generate four types of signals: (i) emission peak shifting; (ii) fluorescence quenching; (iii) fluorescence enhancement/fluorescence turn-on; and (iv) resonance energy transfer (RET), which can generally be defined by the electronic interactions between sensor and VOC. Fluorescent sensors confer high flexibility and multi-variability by providing a variety of parameters for data analysis (e.g., fluorescence intensity, anisotropy, lifetime, emission and excitation spectra, fluorescence decay, and quantum yield).^{133,134} Relying on these features, fluorescent sensors comprising polymers,¹³⁵ small molecules,¹³⁶ and nanomaterials³⁵ have been used for the detection of different types of VOCs. A particular attention was given to the detection of relatively reactive species such as nitramines, nitroaromatics, and peroxide-based explosives at very low concentrations (ppb and ppt).¹³⁷

2.3.3. Optical Fibers and Planar Waveguide-Based Sensors. An optical fiber acts as a waveguide that propagates light by total internal reflection. According to Snell's law, total internal reflection occurs if the incident angle is greater than the critical angle.¹³⁸ When used as VOC sensors, optical fibers can convert the chemical signals into changes in light propagation.^{139–141} Optic fibers are categorized by the transduction process used in the sensing mechanism: (1) fluorescence based, (2) absorption based (colorimetric and spectroscopic), and (3) refractometric.¹²⁷ Generally, optical fibers have short response and recovery times and provide real-time and online monitoring possibilities.¹⁴² Optic fiber-based VOC sensors can be either passive or active. When the sensor response is not linked in any way to an intrinsic change in the optical properties of the fiber, the fiber is considered passive; it acts merely to transport the optical signal to and from the sensing environment. On the other hand, an active optic fiber is modified with receptors/indicators rendering it intrinsically sensitive to VOCs.¹²⁷ Nanomaterials (e.g., graphene^{141,143} and

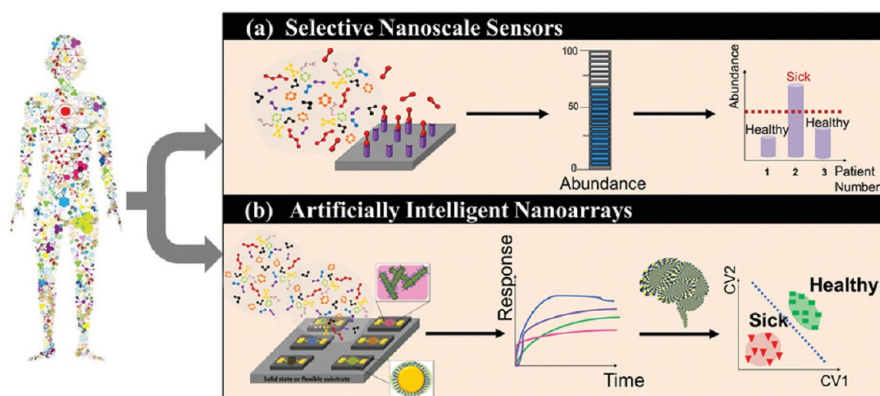


Figure 5. Schematics illustrating different VOC sensing approaches for disease detection: (a) selective sensing approach; (b) cross-reactive sensing approach. Reprinted with permission from ref 9. Copyright 2015 John Wiley & Sons, Inc.

metal oxides¹⁴⁴⁾^{145,146} and polymers^{147–149} have been used as indicators, showing better interactions with VOCs and stronger responses.¹⁴² These modifications are usually achieved by various coating techniques. Dip coating and drop casting methods are extensively used owing to their simplicity and time efficiency. Most of the examples developed during the past few years have been based on unclad fibers coated with sensitive nanomaterials and have proved effective for sensing a wide range of VOC including ethanol, acetone, toluene, chloroform, and formaldehyde.¹⁴² Recently, a cholesteric liquid crystal film–based dual-probe fiber was used to detect various mixtures of ethanol and acetone.¹⁵⁰

Planar waveguide chemical sensors also convert the chemical signal into changes in light propagation.¹⁵¹ These devices provide better compatibility with advanced microfabrication techniques than optical fibers and are suitable for integration with microfluidic and lab-on-a-chip systems owing to their planar geometry.⁴³ Thus, a planar waveguide is an ideal platform for developing integrated optical sensors.¹²⁷ Planar waveguide chemical sensors are also categorized according to the three principal transduction mechanisms in this field: fluorescence, absorption, and refractometry.¹²⁷

2.3.4. Optical Interferometry-Based Sensors. Interferometric VOC sensors measure phase shifts in the superposition of light waves caused by changes in volume/thickness and/or optical properties (e.g., refractive index) resulting from absorption/adsorption of VOCs on a sensing layer (Figure 4).^{152,39} Several optical configurations are used for interferometer sensors. The common techniques used for detecting VOCs are the Fabry–Perot, Sagnac, Mach–Zehnder and Pohl interferometers.¹⁵² Recently, incorporation of nanomaterials into traditional interferometers to enhance sensor performance has received special attention.^{39,152} For example, single-walled carbon nanotubes (SWCNTs) and cadmium arachidate have been used with a fiber optic Fabry–Perot interferometer sensor to detect xylene and ethanol.¹⁵³ The interaction of SWCNTs with electrons in xylene and ethanol vapors resulted in a decrease in free hole density within the SWCNT overlay, leading to changes in their optical properties. These changes were then translated to detectable interferometric signals. Interferometric VOC sensors usually employ optical fibers and waveguides to propagate light to and from the sensing area.

3. VOC TARGETING APPROACHES

Depending on their chemical properties, VOC sensors can probe a range of analyte–sensor interactions to generate a wide range of molecular specificities and responses to different targets. While most interactions are cross-reactive (i.e., less specific and primarily occur when analytes are adsorbed by nonspecific physical interactions), some are more specific and form with a very narrow range of chemicals (ideally a single compound), providing some type of selectivity. These differences have generated two VOC sensing methods: selective vs cross-reactive (Figure 5).

3.1. Selective Sensing. This approach focuses on detecting well-defined and specific VOCs in the presence of interfering gaseous species using a specifically designed and highly selective receptors.¹⁵⁴ It is attractive because the detection can be confirmed while the effect of interfering signals is to some degree neutralized. So far, most selective gas sensors described in the literature have focused on detecting relatively reactive gases such as nitric oxide,^{155,156} hydrogen peroxide,^{157–159} and some VOCs (e.g., acetone).¹⁶⁰ However, selective sensing becomes challenging for less reactive VOCs. This challenge is doubled when the considerable structural, chemical, and electrical similarities of VOCs in complex mixtures in real-world samples is considered. One way to increase selectivity toward specific VOCs is to use very strong sensor–analyte interactions such as covalent and coordination bonds, which are more specific than van der Waals or dipole–dipole interaction. However, this design is associated with very strong binding, which compromises reversibility and the recovery of sensors. A higher working temperature can accelerate recovery but can also lead to irreversible changes in the active sensing layer (e.g., change in the crystalline structure of polymer based sensors).¹⁶¹ Among other methods introduced to enhance selectivity, MIPs have been found very useful due to their ability to interact with specific VOCs, while minimizing the effect of others, based on their 3D structure.¹⁶²

With the continual increase of sensing requirements in the modern era, it is advisable to extend the definition of selectivity of VOC sensors to account not only for chemical sensing but also a wider spectrum of associated signals. In this case, the sensors must be able to respond to VOC signals even when physical conditions vary (e.g., temperature and pressures), a very challenging task that usually requires a combination of multiple sensors and advanced data analysis methods. Additionally, it is important to define selectivity in respect of

the tested sample. For example, selectivity can be defined in a specific subgroup of chemicals and/or between different types of chemicals. Obtaining higher selectivity with the former is more challenging since discrimination among different groups of chemicals (e.g., aldehydes, acids, amines) is usually easier than discrimination among VOCs in the same chemical group (e.g., amines with different aliphatic chains).¹⁶³ In this sense, selectivity should be always defined relatively not absolutely, while providing clear details about the size and composition of the tested sample (i.e., a mixture of a few VOCs compared to breaths that naturally contain thousands of VOCs).

3.2. Cross-Reactive Sensing. When complex mixtures and samples are targeted (e.g., breath and outdoor air samples), the cross-reactive approach is preferable to selective sensors. This alternative strategy mimics the biological system responsible for our sense of smell by using arrays of broadly cross-reactive sensors in conjugation with machine learning (ML).²⁵ This approach is commonly known as “electronic nose” or “artificial nose”.¹⁶⁴ Every sensor in an array responds to all (or a large subset of) the VOCs found in the targeted mixture of compounds.¹⁶⁵ The sensors are sufficiently diverse to provide individually different responses to any given VOC in the mixture but need not be strictly selective. The combined responses of the different sensors are used to establish analyte-specific response patterns (“fingerprints”), usually by classification methods and pattern recognition algorithms.¹⁶⁵ This strategy has many desirable attributes including high detection limits (down to (sub-) ppb), wide dynamic ranges, and low sensitivity toward variable chemical/physical backgrounds. Cross-reactive sensors in the artificial nose format have been successfully used in different applications including disease detection and food monitoring.^{166,167} Even though the current trend of precise odor/molecular recognition is inclined toward artificial nose systems in which sensor arrays replace single sensor units, the selectivity and specificity of a single unit is still highly desired. The optimum sensor array should therefore incorporate a range of sensor elements with a diversity of high specificities toward the targeted analytes and a wide range of chemical interactions.

4. VOC SENSING MATERIALS AND STRUCTURES

4.1. Nanomaterials for VOC Sensing. Nanomaterials with characteristic sizes in the 1–100 nm range have countless advantages in VOC sensors and can be used to overcome many limitations of sensors made from bulk materials.²¹ For VOC sensing, nanomaterials can be categorized as (i) zero-dimensional (e.g., nanoparticles and quantum dots); (ii) one-dimensional (e.g., silicon nanowires (SiNWs) and CNTs); (iii) two-dimensional (e.g., graphene and MXene); and (iv) nonporous 3D (i.e., hierarchical nanoarchitectures).¹¹ Most of these either consist of a single material or comprise several nanometric components (e.g., organic and/or inorganic combinations) so that, in principle, two or more properties that are difficult to achieve in a conventional one-phase material can be combined in the same hybrid nanomaterial system.^{25,27} As such, they have tailorable chemical and physical properties, which relate directly to size, composition, and shape.^{25,168,169} The development of nanomaterials for VOC sensing has been facilitated by progress in nanofabrication and nanotechnology, and found useful for different types of sensors^{27,41} regardless of their transduction methods.

In general, most types of nanomaterials increase sensors' active surface area and interfaces, thus improving sensitivity,

response, and recovery time (Figure 6). In particular, 2D nanomaterials have thicknesses ranging from a few to tens of

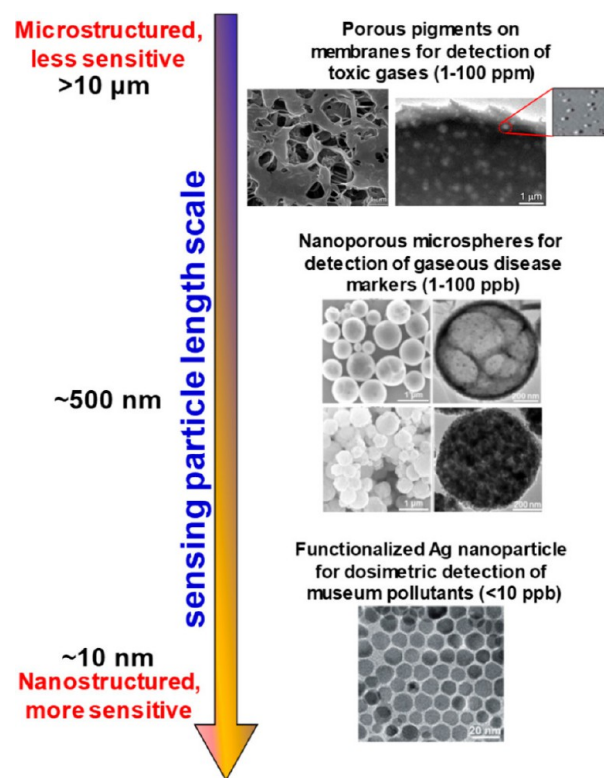


Figure 6. Correlation between sensor micro/nanostructure and detection sensitivity. Reprinted with permission from ref 172. Copyright 2011 Royal Society of Chemistry. Reprinted with permission from ref 173. Copyright 2018 ACS. Reprinted with permission from ref 174. Copyright 2019 Wiley-VCH Verlag GmbH & Co. Reprinted with permission from ref 41. Copyright 2020 ACS.

nanometers and lateral dimensions reaching centimeters, and they have attractive physical and chemical properties for gas sensing.³² Their structures offer a high surface area per unit volume, maximizing the possibilities for material-analyte interactions. Consequently, the properties of these materials are highly sensitive to the chemical environment. Furthermore, the tunable band gap and ability of 2D nanomaterials to act as molecular scaffolds for immobilizing additional recognition components creates an avenue for further improvement and tuning of their sensing properties.¹⁷⁰ In recent years, many types of 2D nanomaterials have been used for VOC sensing including graphene, metal chalcogenides, metal–organic frameworks (MOFs), hexagonal boron nitride (h-BN), black phosphorus, and MXene.

Besides the distinctive chemical and physical properties of nanomaterials, modifying/decorating their surfaces with other nanostructures could improve further selectivity and sensitivity. This has been successfully demonstrated with nanostructured materials such as nanowires, nanotubes, and nanoparticles, and nanometric thin films. Notably, a self-assembled monolayer (SAM) has been shown to be a powerful way of modifying the surfaces of active sensing components because it is easily fabricated, stable, has single layer thickness (faster response), has huge effects on sensing performance, and can be assembled from a wide variety of materials.^{47,53} For example, a series of Si

NW FETs modified by different molecules with different functional groups including phenyl, carboxyl, ester, and alkyl, perfectly recognized 11 kinds of VOCs with the assistance of artificial neural networks.¹⁷¹ Here, the wide variety of nanomaterials generated by surface nanodesign resulted in a highly informative array capable of collecting more extensive information about the environment and its composition than the original materials (i.e., SiNWs).

4.2. Type of VOC Sensing Material. A significant number of nanomaterials have been employed in VOC sensing using various devices and transduction methods. For colorimetric and fluorescent sensors, different types of active materials with chromophore or fluorophore sites have been introduced for converting the chemical signal. These include Lewis acid/base dyes, Brønsted acid/base dyes, redox indicator dyes, and solvatochromic or vapochromic dyes.³⁷ For electrical sensors, the variety of nanomaterials is even greater.¹⁰ It includes organic, inorganic, and hybrid sensing materials such as carbon materials, polymers and their composites, and metal oxides. Each subtype has its own advantages and limitations. The variety of sensing nanomaterials has been increased by the enormous possibilities for chemical modifications, hybridization, and structural engineering, which have fueled the development of powerful cross-reactive sensor arrays with multiple recognition sites. The following section discusses different categories of sensing materials that have been commonly used for VOC sensing. The different categories are not necessarily mutually exclusive as they show some overlaps.

4.2.1. Metalloids, Metals, Metal Oxides, and other Inorganic Compounds. This group includes a variety of inorganic materials used for VOC sensing. Metal oxides were among those materials to be used for VOC sensors and to be commercialized successfully. They can be divided into two groups according to their electronic structures: transition metal oxides (e.g., NiO, Cr₂O₃) and nontransition metal oxides (e.g., ZnO, Al₂O₃).¹⁷⁵ Transition-metal oxides have more than one preferred oxidation state and are better suited for gas sensing.¹⁷⁵ For sensitive detection at ppb levels, binary, ternary, quaternary or noble metal-decorated metal oxides are usually used.¹⁷⁵ Metal oxides are mostly used in chemiresistors as sensing layers but can also be used in FET devices,^{160,176} and usually operate at higher temperatures than other sensors.⁷⁴ For example, an indium nitride FET gas sensor with a 10 nm-thick active layer and a Pt catalyst achieved a detection limit of 0.4 ppm acetone under 200 °C.¹⁶⁰

In general, metal oxide layers have a high affinity for negatively charged oxygen moieties (e.g., O₂⁻, O⁻), which leads to the creation of either an electron-depletion or a hole-accumulation surface charge layer according to the type of semiconductor, n-type and p-type, respectively.¹⁷⁷ The gases react with these oxygen moieties inducing variations in the space charge layer of surface-trapped electrons, which leads to a change in device resistance.¹⁷⁷ Multiple parameters affect the responses of metal oxides to VOCs. These include composition, size, shape, morphology, surface area, fabrication method, doping level, microstructure of sensing layers, humidity, and operating temperature. Tuning these parameters can provide highly selective and sensitive control over the detection of interesting gases under real-world conditions.^{74,178–180} In one study, Mg-doped In₂O₃ nanowire FET sensor arrays decorated with various discrete metal nanoparticles (Au, Ag, and Pt) were used to detect different

gases.¹⁸¹ The sensors operated at room temperature and distinguished unequivocally among three reducing gases (CO, H₂, and C₂H₅OH), illustrating the value of metal nanoparticle catalysts for tuning the sensing performance of metal oxide-based devices.

SiNWs are important members of in this group of sensing materials. SiNWs are excellent candidates for gas sensing applications because they combine several advantages including large carrier mobility,¹⁸² tunable properties controlled by doping levels,¹⁸³ and/or chemical functionalization.¹⁸⁴ Importantly, SiNW-based gas sensing applications are highly compatible with conventional electronic systems, which should accelerate their integration into real-world applications.⁹⁴ SiNWs are mainly used in FET devices owing to their excellent semiconducting properties, but have also been used in chemiresistors.⁷¹ Bare SiNWs can interact with very reactive gases and polar VOCs but have minimal interaction with nonpolar VOCs, which is a challenge for sensing. In 2011, Niskanen et al.¹⁸⁵ reported an unmodified SiNW that can be used in combination with pattern recognition algorithms to produce versatile chemical vapor sensors without additional functionalization. They demonstrated 100% accuracy in detecting acetone, ethanol, and water and 96% accuracy for methanol, ethanol, and 2-propanol under ambient conditions. Fortunately, the sensing performance of SiNWs can be tuned in different ways. Doping with different atoms and surface decoration with nanoparticles and organic compounds have proved successful.⁹³ The fundamentals and applications of functionalized SiNW-based FETs for detecting both polar and nonpolar VOCs were systematically studied by Haick et al.^{51,90,171,186–191} In one study, they reported the sensing properties of SiNW functionalized with hexyltrichlorosilane (HTS) to both polar (water, ethanol, 1-butanol, 1-hexanol, 1-octanol and 1-decanol) and nonpolar (*n*-hexane, *n*-octane and *n*-decane) VOCs at ppm levels in 15% RH air.¹⁸⁹ HTS modification greatly enhanced the sensitivity to both polar and nonpolar VOCs. In a subsequent study, Wang et al.^{51,90} investigated the effect of chain length and functional groups on the sensing properties of SiNWs to various VOCs. The results have shown that functionalization of SiNW FETs with longer monolayers of alkane-backbone silane or with functional groups that induce more effectively dipole-dipole interaction between the molecular layers and nonpolar VOCs made the sensors more selective.

Metal nanostructures modified with organic molecules have been widely used for VOC sensing because they have properties such as large surface-to-volume ratios, room temperature operation, low voltage operation, and fast response and recovery times.^{192–194} Monolayer-capped gold nanoparticles (GNPs) have received particular attention.^{52,195–198} They are usually used in chemiresistor devices that can be fabricated by various simple methods including drop-casting,^{26,199,200} layer-by-layer (LbL) deposition,^{201–203} spray coating,²⁰⁴ spin-casting,⁶¹ printing,^{205,206} and others^{207–209} on both solid or flexible substrates. The fabrication method has a critical effect on the sensor properties. Bashouti et al.²⁰³ compared the performances of sensors prepared by two fabrication methods, drop casting and LbL. LbL gave more homogeneous films than drop-casting. The sensors were tested under exposure to polar and nonpolar VOCs associated with health monitoring, including hexane, ethylbenzene, water, and ethanol. Interestingly, sensors

prepared by drop-casting were more sensitive than those prepared by the other method.

Other types of VOC-sensing nanomaterials belong to this group such as chalcogenides,²¹⁰ tellurium,^{211,212} MXenes,²¹³ and black phosphorus.^{46,214} Some of these materials are relatively new and/or poorly studied in VOC sensing, so they require more exploration and comparison to other types of VOC sensing materials.

4.2.2. Carbon Materials. Carbon nanomaterials have inherent nanoscale features and can potentially become ideal components for the next generation of VOC sensor technology. They combine excellent detection sensitivity with interesting transduction properties in a single layer of material. Carbon materials include CNTs, carbon black, graphene, and carbon fibers. Low-dimensional carbon structures have most of their atoms exposed to the environment, so they offer a high specific surface area, which is advantageous for optimizing sensitivity. CNTs and graphene specifically have high-quality crystal lattices and show high carrier mobility and low noise. The sensitivity and selectivity of carbon nanomaterials can be engineered by different techniques both to create defects and to graft functional groups on their surfaces in a controlled way.²⁸

Carbon black particles are the oldest materials used for VOC sensors.²¹⁵ Usually, they are dispersed into insulating organic polymers. Carbon black makes the films electrically conductive, whereas the organic polymers provide chemical diversity between the sensor elements. Swelling of the polymer upon exposure to a gas/vapor increases the resistance of the film, providing an extraordinarily simple way of monitoring VOCs.^{215–219} The relationship between resistivity and carbon black content in carbon black–organic polymer composites is well described by percolation theory.²²⁰ At low carbon black loadings, the composites are insulators because there is no connected pathway of conductive particles across the material. As the carbon black content increases, there is a sharp transition in which the resistivity of the composite decreases dramatically over a small difference in carbon black concentration. At this transition point, the percolation threshold, a connected pathway of carbon black particles is formed. When this composite film swells by absorbing gases, the conduction pathway is disrupted, leading to changes in resistance. Lewis and co-workers employed commercially available carbon black and different polymer compositions in each sensor element.^{221,222} The hybrid structures comprising carbon black dispersed in polymer matrices were used as arrays of sensing elements for identifying different VOCs including benzene and methanol. The response was moderate considering the extremely high concentrations tested. Huynh et al.²¹⁷ also used carbon black in advanced gas sensors with self-healing capabilities.

It has to be noted that carbon black–polymer composite sensors can suffer from baseline and response drift owing both to aging of the polymer matrix and to rearrangements of carbon black particles within the polymer, leading to changes in the percolation paths. These rearrangements occur progressively because of the many swelling/shrinking processes upon repeated detection and recovery cycles.²⁸

CNTs are another promising type of nanomaterial for a wide range of VOC detection applications. They can be divided into two groups according to their structure: single-walled CNTs and multi-walled (MWCNTs). CNTs attract the attention of many researchers mostly because of their size range, large

surface-area-to-volume ratio, electrical and mechanical properties, and compatibility with other nanomaterials for generating enhanced properties. CNT diameters are in the range 0.8–2 nm for SWCNTs and 5–100 nm for MWCNTs, and the lengths of vertically aligned CNTs range from microscale to milliscale.²²³ MWCNTs show metallic properties whereas SWCNTs show either metallic or semiconducting behavior depending on the chiral angle between the hexagons and the tube axis. Semiconductive CNTs have excellent electrical properties such as high field effect mobility.^{224,225} For electrical sensing of VOCs, CNTs can be used in either FET or chemiresistive devices.^{10,223} They have superior mechanical properties, making them excellent candidates for durable flexible gas sensors.^{226,227}

The electronic properties of CNTs are extremely sensitive to their local chemical environment. This sensitivity has made them ideal candidates for incorporation into the design of chemical sensors.²⁸ Many studies have shown that functionalization of CNT sidewalls enables a better chemical bonding with specific chemical species to be designed, enhancing the selectivity of adsorption (Figure 7).^{28,228,229} For example,

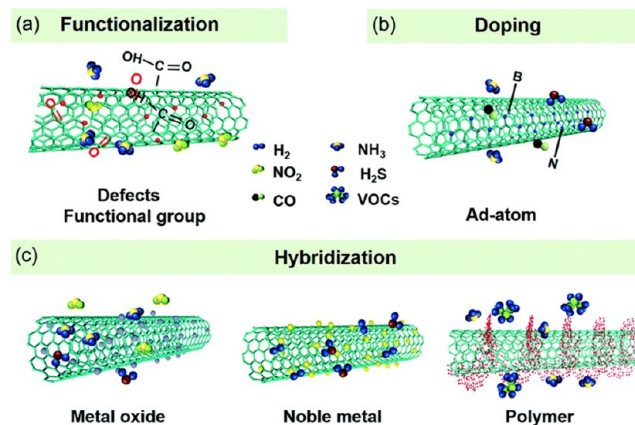


Figure 7. Schematic illustration of strategies to promote sensing by CNTs via doping, surface functionalization, and hybridization. Reprinted with permission from ref 56. Copyright 2020 Royal Society of Chemistry.

Penza et al. introduced the use of sputtering as a way to decorate CNTs with metal nanoparticles (e.g., Au, Pt, Ru, or Ag),²³⁰ which were then used to detect a few inorganic gases and methane. CNTs can also be decorated with organic molecules such as DNA to impart selectivity.²³¹ In one study, DNA-coated CNT devices responded to gas odors (e.g., explosives or nerve agent simulants) that caused no detectable conductivity change in bare CNTs. Their selectivity could be tuned in respect of the sign and magnitude of response by choosing the base sequence of the single-stranded DNA. In another demonstration, the sensing properties of hybrid CNT–porphyrin films were reported.²³² The authors explored the possibility of transducing adsorption events in a porphyrin layer into resistance changes in the CNT layers. The porphyrin films increased the sensitivity of the electric resistance of the CNTs to VOC concentrations, probably because the porphyrin was efficient in conveying the charge transfer from the adsorbate molecule to the CNT. The authors suggested that this property could be further exploited by using different metalloporphyrin–CNTs in sensor array configurations to adjust selectivity. Star's group employed a nature-inspired CuI

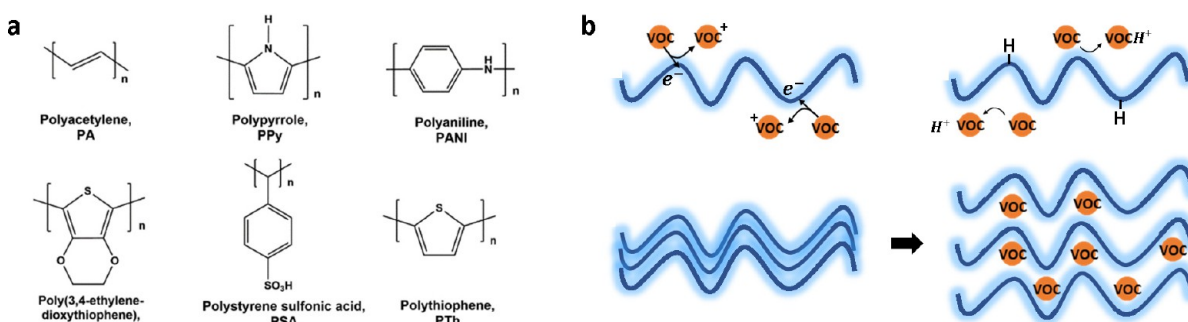


Figure 8. (a) Examples of representative conductive polymers used for VOC sensing. Reprinted with permission from ref 56. Copyright 2020 Royal Society of Chemistry. (b) Schematic illustration of sensing mechanisms for conducting polymers via charge doping, deprotonation, and swelling.⁵⁶

complex to fabricate single-walled CNT sensors that could selectively detect 0.5 ppm of ethylene; this can be used to monitor the ripening of climacteric fruits.²³³

The poor solubility of CNTs hinders their chemical functionalization and the subsequent integration of these materials into devices. Swager et al. described a solvent-free procedure for rapid prototyping of selective CNT chemiresistors on paper. This procedure enables functional gas sensors to be fabricated from commercially available starting materials in less than 15 min. The first step involved the preparation of solid composites of CNTs with small molecules (i.e., selectors, designed to interact with specific classes of gaseous analytes) by solvent-free mechanical mixing and subsequent compression. The second step involved the deposition of chemiresistive sensors by mechanical abrasion of these solid composites on to the paper surface. These materials generated cross-reactive arrays capable of sensing and differentiating multiple gases and VOCs.

The use of CNTs has not been limited to electrical sensors. Penza et al.^{234,235} investigated SAW sensors coated with either SWCNTs or MWCNTs for detecting different VOCs (ethanol, ethyl acetate, and toluene) at room temperature. Interestingly, the selectivity to VOCs was controlled by the type of organic solvent used to disperse the CNTs and the number of CNT layers. The highest sensitivity to ethanol was obtained with SWCNTs in ethanol, whereas for the other VOCs it was obtained with SWCNTs in toluene. SWCNTs have also been applied with optical fiber-based sensors because they can translate chemical interactions into changes in optical properties.¹⁵³

Graphene is an allotrope of carbon with a two-dimensional atomic layer arrangement of sp^2 -hybridized carbon atoms covalently linked in a honeycomb lattice. Its extraordinary electrical, physiochemical and structural properties^{236–242} such as very high carrier mobility and carrier density, very large surface area of $2630\text{ m}^2\text{ g}^{-1}$, mechanical strength ($>1060\text{ GPa}$), and intrinsically low electrical noise make it one of the most promising materials for gas sensing.^{243–246} It can now be produced in bulk quantities by chemical vapor deposition (CVD) methods,^{247–249} CNT unzipping,^{250,251} and most efficiently by the oxidation and exfoliation of graphite, which is then reduced to reduced graphene oxide (rGO), a derivative of graphene.^{252–255} Graphene and rGO have been extensively reported as sensing materials for FETs, chemiresistors and other sensors in both solid state and flexible designs.^{237,238,256}

By controlling graphene production and its modification, its sensing of VOCs can be highly tuned.²⁵⁷ Introducing defects

and dopants into graphene sheets can alter VOC sensing drastically.^{236,258} Theoretical studies have shown that adsorption of VOCs is improved when graphene is doped.^{259,260} An et al.²⁶⁰ showed that an Al-decorated porous graphene monolayer enhances the adsorption energies of carbonyl-containing VOCs nearly 2-fold, though there were only slight changes for other VOCs. These results provide a basis for preparing sensors with high selectivity toward carbonyl-containing VOCs. Many experimental studies have further proved the effect of dopants and defects in tuning the chemical sensitivity.^{236,240} For example, the type and geometry of graphene defects affect the response to VOCs.²⁶¹ Pristine graphene with only a few point defects was insensitive to 1,2-dichlorobenzene and toluene. However, when line defects were introduced, the sensitivity increased. The authors continued by cutting the graphene sheets into $2\text{--}5\text{ }\mu\text{m}$ wide ribbons. This increased the response to 1,2-dichlorobenzene to double that of an unpatterned CVD-graphene.

The use of rGO in chemical sensors has attracted more attention than other graphene derivatives, mostly because of its availability at relatively low cost and in large scale production,²⁵² the abundant defects and chemical groups in the rGO structure that facilitate gas adsorption,²⁶² and its highly tunable chemical and electrical properties.²³⁸ Incomplete chemical or thermal reduction of GO is effective for partially restoring conductivity while retaining some oxygen-containing groups (e.g., hydroxyl, epoxy, carbonyl and carboxyl) as defects, which are useful for sensing. The reduction process can also introduce voids and structural defects that act as further adsorption sites.²⁶³ Optimization of defect density is an effective way of balancing the sensitivity and recovery rate of a rGO-based gas sensor. Another highly promising way of improving the sensitivity and selectivity of graphene-based chemical sensors is via functionalization and hybridization.^{56,238,264,265} These methods offer enormous numbers of graphene-based sensing material combinations with tuned chemical reactivity, such as graphene containing metal and metal oxide nanoparticles, polymers, and small organic molecules.^{265–268} Besides the straightforward use of graphene in electrical sensors, its properties are useful for other VOC sensing devices including colorimetric,²⁶⁹ optical fiber,¹⁴¹ and QCM sensors.^{270,271} In these examples, graphene (or its derivatives) succeeded because of its high surface area and VOC absorption capabilities, not its electrical conductivity. Notably, Cao et al.²⁶⁹ reported the use of a polydiacetylene (PDA)–graphene hybrid for colorimetric VOC sensing. Here, graphene efficiently supported a self-assembled PDA film

owing to its large surface-to-volume ratio and high transparency, enhancing the sensitivity of the PDA sensor. The hybrid colorimetric sensors were tested under exposure to four different VOCs: tetrahydrofuran, CHCl_3 , CH_3OH , and dimethylformamide. All showed higher responses than pure PDA.

Other types of carbon materials such as carbon quantum dots have been involved in VOC sensing applications.^{272–275} They are highly promising owing to their fluorescent properties, which facilitates their integration into optical VOC sensing platforms.

4.2.3. Polymers. Polymers are attractive candidates for designing and decorating different types of VOC sensors. Different forms of polymers have been used for this purpose: (i) conductive and semiconductive polymers in electrical sensors that constitute the active sensing layer; (ii) colorimetric and fluorometric polymers; (iii) polymer composites and hybrids, including both insulating and conductive polymers, in different configurations of sensing devices (e.g., gravimetric, optical, and electrical).

Conducting polymers are particularly attractive for detecting VOCs.^{30,276} Polypyrrole, polyaniline (PANI), polythiophene, and their derivatives have been used as active layers of gas sensors since the early 1980s (Figure 8a).⁷⁰ The polymeric molecular backbone and side chains/groups can easily be tailored, producing variations in electrical properties and surface chemistry and eventually a tunable response toward specific VOCs.^{31,277} The huge range of possibilities for modification allows polymers to be designed and selected rationally for developing very powerful sensor arrays.^{278,279} Most importantly, conducting polymers are easy to synthesize by chemical or electrochemical processes, have good mechanical properties, are solution-processable, and can be fabricated using various printing techniques.^{56,280} Polymer films can be deposited using spin coating, spray coating, electrochemical deposition, dip coating, layer-by-layer, Langmuir–Blodgett technique, thermal deposition, and other methods.

When polymers are used for VOC sensing, gaseous analytes can either react chemically or interact with them via weak intermolecular forces such as π – π interaction, H-bonding, and dipole–dipole interaction, depending on their chemical activities (Figure 8b).^{281,282} These interactions change the electrical conductivity of the polymer. Several mechanisms have been suggested to explain this transduction. It is now accepted that some VOCs change the electrical properties of the conducting polymer film via gas-induced charge doping.⁵⁶ This was initially suggested for some inorganic gases such as NO_2 and NH_3 . In view of the Lewis acid and base characteristics of these compounds, the transduction was explained by “secondary doping”, whereby electron donation or withdrawal by the vapor changed the conductivity. A similar mechanism was suggested to the more common VOCs, leading to a model of partial charge transfer between polymer and analyte, although the Lewis acid and base properties of these compounds are much less prominent.²⁸³ Protonation/deprotonation can also explain why polymer conductivities change upon exposure to some VOCs.⁵⁶ For example, deprotonation of imine sites governs the response of PANI under exposure to butylamine; the conducting emeraldine salt forms are converted to insulating emeraldine base forms, decreasing conductivity.²⁸⁴ Other VOCs interact only weakly with polymers, which is important for detecting organic analytes

with very low reactivity such as benzene and toluene and other nonpolar VOCs. Given the strong affinity of most VOCs for polymers via various intermolecular interactions and the abundant adsorption sites on polymeric chains, polymer films can take up large numbers of VOC molecules. This causes structural changes that can also modify the electrical properties. For example, VOC-induced swelling of conductive polymers enlarges their interchain distance, which restricts electron hopping between chains and reduces the conductivity of the sensing film (Figure 8b).²⁸⁵ In contrast, in very specific cases, certain alcohols and ketones can interact strongly with the nitrogen atoms in a conducting polymer such as PANI, leading to more stretched conformations and improved electron transport.²⁸⁶ Major advantages of PANI for sensing devices are its ease of synthesis, environmental stability, acid/base doping/dedoping, and oxidation/reduction property. Moreover, PANI can be shaped into various structures with different morphologies, making it possible to obtain nanofibers as well as thin films with improved processability and functionality.^{287,288}

Semiconductive polymers have also gained attention for VOC sensing in OFET devices. They combine the advantages of conducting polymers with the multiparametric operation of FETs. For example, Wang et al. demonstrated the multiparametric behavior of FET-based VOC sensing using an ambipolar semiconducting polymer, poly-(diketopyrrolopyrrole-terthiophene) (PDPPHDT-3), to detect xylene isomer vapors.⁸⁹ The ambipolar FET generates multiple sensing parameters through electron and hole conduction. Using a combined pattern recognition method along with the multiple sensing parameters obtained from a single ambipolar OFET, chemically and structurally similar xylene isomers could be identified, as well as their binary and ternary mixtures. In a subsequent study, a wider range of compounds including xylenes, ketones and aldehydes was identified and successfully distinguished by a high-mobility polymer FET using both the device's mobility and threshold voltage as the sensing parameters.²⁸⁹

Polymers have also been used in other types of sensors. The extensive uptake of VOC molecules by a polymer film can increase its mass, facilitating its use in gravimetric sensors. For example, by recording the mass change of a poly-(etherurethane) film with a QCM, different kinds of VOCs including ethanol and acetone were detected.²⁹⁰ Colorimetric and fluorescent polymers have also been used to prepare optical sensors (see section 4.2.4).

As already mentioned, polymers have also been used for VOC sensing in complex forms such as MIPs and porous polymeric membranes, which increase the surface area of the active layer and impart shape-based selectivity, improving the general performance of sensor arrays. MIPs are prepared by the cross-linking monomers around the target analyte to create a polymer film template with binding sites that perfectly match that analyte.^{291–293} MIPs can be extremely selective and overcome some of the pronounced disadvantages of pure polymers, especially the slow response times.¹⁶² However, unlike many pure polymers or other commercially available sensitizers, they require individual chemical preparation because they must be imprinted for a specific analyte. There are many different techniques for successfully imprinting polymers, as addressed in other review papers.^{293,294} MIPs have frequently been combined in all types of VOC sensors including electrical, optical, and gravimetric devices.

4.2.4. Materials for Optical Sensing. Colorimetric and fluorometric nanomaterials are widely used in optical VOC sensors. They rely on functional sites to interact with analytes and a chromophore or fluorophore to couple with the active site.³⁷ The types of intermolecular interactions that can induce significant colorimetric or fluorometric changes suggest five classes of cross-reactive, chemoresponsive dyes (with some overlap): (i) Brønsted acidic or basic dyes (e.g., various pH indicators); (ii) Lewis acid/base dyes (e.g., metal complexes with open coordination sites or metal-ion-containing chromogens); (iii) redox dyes; (iv) colorants with large permanent dipoles (i.e., zwitterionic vapochromic or solvatochromic dyes) for detecting local polarity or hydrogen bonding; and (v) chromogenic aggregative materials (e.g., plasmonic nanoparticles and nanoscale transition metal sulfides). This variety of interactions provides multiple design opportunities for controlling and tuning the performances of colorimetric and fluorometric sensors for specific applications.³⁷

Porphyrins and metalloporphyrins have a ring-based structure that is the basis of many pigments such as hemoglobin and chlorophyll.¹⁶⁶ These nanomaterials have received wide attention in colorimetric VOC sensing.³⁷ They are stable and well-characterized and can be modified by changing either the components of their ring structure or the metal atom bound in the center.²⁹⁵ Suslick et al. have used porphyrins and metalloporphyrins, in addition to other colorimetric materials, to develop the early versions of colorimetric sensor arrays.^{131,163} Exposing those arrays to vapors of chemical compounds led to interaction with the dyes and generated distinct color changes that could be recorded with a flatbed scanner. In more recent work, these materials have been parts of upgraded arrays that include a wider range of chemical dyes.²⁹⁶ In one example, the sensor array contained 40 elements and was successfully used for detecting toxic or explosive vapors (Figure 9).^{296,297} For more

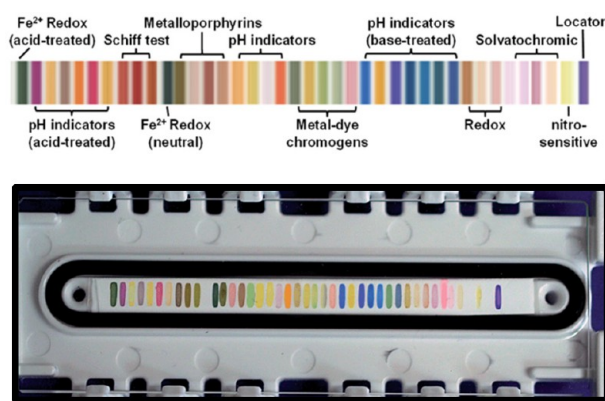


Figure 9. Example of a VOC sensor array made of 40 colorimetric dyes. Reprinted with permission under a Creative Commons Attribution 3.0 Unported (CC BY 3.0) License from ref 130. Copyright 2016 Royal Society of Chemistry.

information about the types of sensing materials used in this array, please refer to another review.³⁷ One big advantage of these colorimetric arrays is that they can be made insensitive to humidity. However, they might be restricted to single use because most of the studied vapor-sensor interactions are irreversible.

Fluorometric and colorimetric polymers have also been used for VOC sensing.^{298,299} PDAs in particular have been

extensively studied owing to their structural, spectral, and optical features.²⁹⁸ The colorimetric and fluorescent transitions of PDAs in response to different external stimuli (including VOCs) make them good sensing materials. As with other conjugated polymers, responsivity toward VOCs can be tuned by designing the main backbone and side groups, providing a large structural diversity of PDAs.^{300–303}

Plasmonic sensing nanomaterials are also promising for VOC optical sensing applications.³⁰⁴ The sensing mechanism is based primarily on the optical properties associated with localized surface plasmon resonance (LSPR, the resonance oscillation of free electrons at surface boundaries in metal films) on metal nanoparticles and nanostructured surfaces.³⁰⁵ The enhanced electromagnetic fields at the surface of a resonant plasmonic structure allow variations in the environment to be probed.³⁰⁴ The working mechanism of surface plasmon resonance (SPR) VOC sensors depends on the change of refractive index caused by the capture of VOC molecules, which shifts the resonance wavelength.³⁰⁶ Many plasmonic-based VOC sensors have been described in the literature. The most widely employed plasmonic structures involve Au and Ag. Au is the most common material because it is chemically inert and stable. Ag is postulated to have superior plasmonic properties but is sensitive to oxidation under ambient conditions. In one example, silver nanoparticles were used in an SPR-based system to detect toluene, octane, chlorobenzene, *m*-xylene, and pentanol.³⁰⁵ Upon exposure, the VOCs were adsorbed on the metal surface and formed a thin organic layer, shifting the LSPR wavelength of the Ag nanoparticles. This study exemplified reagent-free sensors in which the VOC interacted directly with the plasmonic entity. This type of sensing is usually difficult owing to the relatively low adsorption efficiency of VOCs on plasmonic metals and the small change in refractive index per adsorbed molecule. As an alternative, Chen and Lu suggested using Ag nanoparticles modified with a thiol-based monolayer for imparting selectivity toward such VOCs as heptanone, butanol, xylene, octane, butyl acetate, and propanol.³⁰⁷ In this system, when the VOC is adsorbed on a modified Ag nanoparticle, the thickness of the organic layer on the Ag surface increases. Moreover, some vapor molecules such as alcohols penetrate the thiolate monolayer and adsorb on to the metal surface. Both phenomena reduce the volumetric percentage of air in the localized surface plasmon resonance sensitive region and shift the LSPR spectrum of the nanoparticles. SAM modification not only alter the chemical affinity of the surface, but also improves the detection limit without affecting the response time. So far, advances in this area have allowed the size, shape, and surface chemistry/modifications of plasmonic nanomaterials to be precisely controlled, thus providing opportunities to design sensors for particular VOCs. Many types of materials have also been used to nanodecorate plasmonic nanostructures and to provide control over sensitivity and selectivity toward VOCs. These include polymers,³⁰⁸ MOFs,^{309,310} and carbon materials.^{311,312} Plasmonic nanomaterials have proved very useful in combination with optic fibers and waveguides.³¹³ Because of their high optical activity, they can be used as receptors on optic fibers to probe changes in the chemical environment. Here, the chemical signals are interrogated via the measured loss in transmitted/reflected light at the resonance wavelength.

Photonic crystals consist of periodic arrangements of materials with different dielectric constants. They have distinct

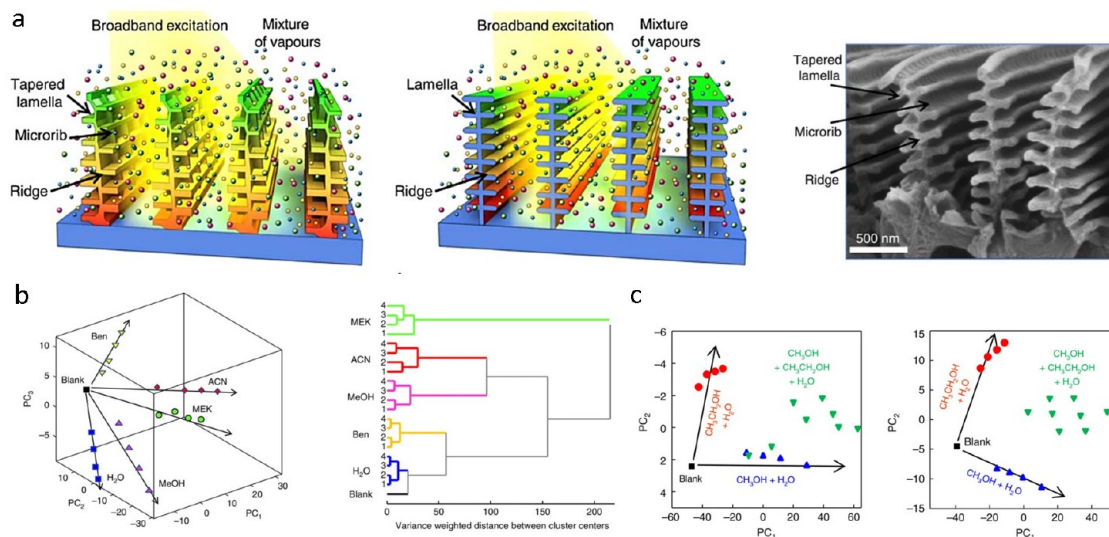


Figure 10. (a) Schematic of the tree-like tapered structure of natural butterfly scales and similar synthetic nanostructure used for VOC sensing. (b) Discrimination and classification of five vapors at four concentrations using PCA and hierarchical cluster analysis. (c) Responses to methanol and ethanol in the presence of water using bare and functionalized nanostructures. Reprinted with permission under a Creative Commons Attribution CC BY License from ref 317. Copyright 2015 Springer Nature.

wavelengths of reflection governed by their periodic structure, which provide their specific color. Changes in the periodicity of the crystal, caused for example by a chemical stimulus, alter the wavelength of maximum reflectance of the photonic crystal; this alteration can be used for sensing purposes.^{314–316} Photonic crystals can be used in different sensor configurations such as on-chip nanoscale resonators or waveguides, Bragg stacks made of mesoporous crystals, and colloidally templated 3D crystals.³¹⁴ Deng et al.³¹⁷ developed high-selectivity vapor sensing using Morpho scales as a bioinspiration. This sensor comprises a multilayer interferometric nanostructure with individual horizontal lamellae supported through their middle by a vertical ridge, a resonant structure designed to promote a greater vapor response than nonresonant sensors (Figure 10a). Moreover, the periodic arrangement of these nanostructures adds diffractive effects to the sensor response. The 3D nanostructure-based sensors can distinguish individual closely related vapors in dry-gas conditions and even in mixtures with a variable moisture background (Figure 10 b,c). The authors also suggested design criteria to tune the selectivity of sensors, including spatial orientation of surface functionalization, chemistry of surface functionalization, and extinction and scattering of nanostructure.

4.2.5. Small Molecules and Metal Organic Frameworks. Small molecules constitute a wide group of VOC sensing materials, both as receptors and as active sensing materials. They offer multiple advantage such as easy processability and numerous chemical modification possibilities. We have already mentioned some examples of small molecules in previous sections, such as metal porphyrins, pH indicators, and other fluorometric and colorimetric small dyes. These materials have been primarily used in optical sensors as the main active components or as receptors in electrical and gravimetric devices. Small molecules with semiconductive properties have also been employed in VOC sensors.^{99,318,319} These molecules are usually thermally evaporated to obtain thin semiconductive sensing films in FET devices.³¹⁸ Many examples of FET VOC sensors based on semiconductors such a pentacene and copper phthalocyanine derivatives have been shown in the liter-

ature.^{320–323} Kang et al.³²² reported the development of a high-performance pentacene-based OFET methanol sensor. The device was prepared by sequential evaporation of *m*-bis(triphenylsilyl) benzene (TSB3) as a small-molecule dielectric layer and pentacene as the active material on an *n*-octadecyltrimethoxysilane (OTS)-treated SiO₂ substrate. This produced a macroporous pentacene/TSB3 layer that showed high sensitivity toward methanol vapor. Besides their conventional role in thin film transistors, semiconductive small molecules have also been used as receptors in other sensing devices such as gravimetric sensors and chemiresistors.^{40,324}

MOFs, a class of crystalline and porous materials formed by molecular self-assembly of metal-containing nodes with organic linkers (e.g., carboxylate, hydroxyl, thiol, and amino) to form bulk crystals, have been considered promising candidates for chemical sensing of VOCs. MOFs have many advantageous properties such as large porosities, highly specific surfaces, and highly tailorable structures.³²⁵ The rational tuning of MOF pore sizes allows for the selective adsorption and separation of small VOC molecules through size-exclusion effects, enabling the selective detection of specific targets. MOFs can easily assemble into uniform films due to their crystalline nature, which is very useful for fabricating sensing devices.³²⁶ Owing to their structural and functional tunability, MOFs have been successfully applied in multiple types of VOC sensing devices including electrical, colorimetric and fluorometric, and gravimetric.³²⁵

MOFs are important examples of colorimetric and luminescent sensing nanomaterials.^{325–328} The luminescent properties of MOFs are by far the most widely explored for sensing to date. Luminescent-based sensing in MOFs has been primarily based on quenching, and occasionally enhancement, of photoinduced emission due to VOC adsorption.³²⁵ Latham et al.³²⁷ developed an infinite 1D ribbon MOF based on a network of tetrapyridinetetraphenylethene molecules coordinating with ZnCl₂ nodes, which exhibited a strong aggregation-enhanced fluorescence emission. This network exhibited enhanced turn-on fluorescence in response to methyl-

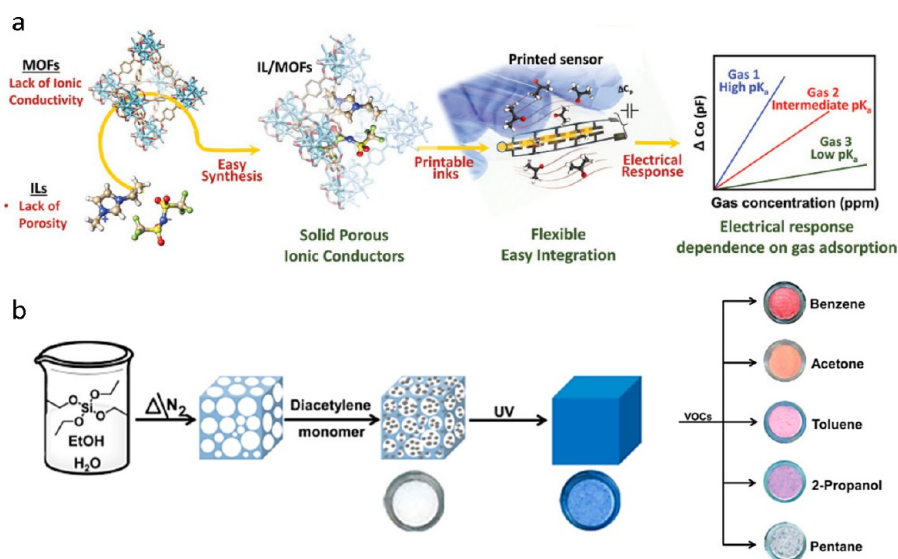


Figure 11. (a) Schematic illustration showing the ionic liquid/MOF hybrid preparation and its utilization as printed sensor for gas detection. Reprinted with permission from ref 338. Copyright 2021 John Wiley & Sons, Inc. (b) Preparation of the PDA-aerogel and its colorimetric sensing properties. Reprinted with permission from ref 336. Copyright 2019 ACS.

substituted VOCs and quenching in response to nitro-substituted VOCs.

The colorimetric sensing properties of MOFs rely on the phenomena of solvatochromism and vapochromism, which refer to the shift in the absorption spectrum of a material in response to exposure to solvents and vapors. For example, a dihydrotetrazine-functionalized MOF was used to prepare a colorimetric gas sensor for detecting chloroform.³²⁹ This MOF changes color from yellow to pink in the presence of chloroform owing to the reversible dynamic conversion of dihydrotetrazine to tetrazine. It has a short response time (10 s) and can be used to detect chloroform selectively in the presence of other VOCs.

It is important to note that the use of MOFs in VOC and chemical sensing has not been limited to luminescent and colorimetric methods. MOFs have also been used in electrochemical, interferometric, localized SPR, colloidal crystal, and electromechanical (e.g., QCM and SAW) sensors and impedance spectroscopy-based methods.^{325,330,331} Chen et al. has demonstrated an in situ self-assembly of MOFs on a SiO₂ waveguide for the development of asymmetric Mach-Zehnder interferometers-based VOC sensor. This device showed high-performance ethanol sensing with a large detection range (0 to 1000 ppm), high sensitivity (19 pm ppm⁻¹ from 0 to 50 ppm or 41 pm ppm⁻¹ from 600 to 1000 ppm), and a low detection limit (1.6 ppm or 740 ppb). In another example, Achmann et al.³³² has demonstrated the application of three different commercially available MOF materials: Al-terephthalate-MOF (Al-BDC), Fe-1,3,5-benzenetricarboxylate (Fe-BTC), and Cu-1,3,5-benzenetricarboxylate (Cu-BTC) in impedance spectrometric sensing devices. The impedance spectra exhibited measurable changes as the MOF was exposed to various VOCs such as C₃H₈, ethanol, and methanol.

4.2.6. Hybrid Materials. The intimate combination of two or more materials on the nanometric scale has provided an alternative way for developing VOC sensing devices.^{333–337} Hybrids can be obtained by combining materials differing in nature such as inorganic metal oxide, metallic nanoparticles, or organic components such as conducting polymers, graphene

and CNTs for optimizing the final properties. Hybrid nanomaterials can be obtained through a broad range of methods, depending on the intended application and properties. The nature of the constituent materials, the ratios between them and the synergistic interaction at the shared interface are key to design improvement.²⁷ Careful design of hybrid composites can improve the sensing of target VOCs over the individual constituents (e.g., higher selectivity and sensitivity). This method has proved useful for electrical, gravimetric, and optical VOC sensors.

This strategy for chemiresistors has recently been reviewed.²⁴ The authors have distinguished five typical hybridizing forms that can be used for improved sensing characteristics. The first combination relies on catalyzed reactions (normally noble metal catalysts, e.g., Pt, Pd, Au, Ag) between analyte gas and decorated catalysts on the host semiconductive materials. The second relies on a fast charge transfer process, viz. electrical carrier withdrawal or donation between guest additives and the host material (e.g., CNTs, rGO). The third relies on regulating the charge carrier transport in conductive/semiconductive materials upon exposure to gas analytes. The fourth relies on manipulation/construction of the heterojunctions such as n–n, p–n, p–p, p–n–p between heterogeneous semiconductive materials. The last relies on semiconductors coated with gas molecular sieving/binding layers or ligands/complexes for selective gas detection. Despite useful discussion, there remains a need for papers focusing on how the design of hybrid nanomaterials can be used to optimize VOC sensors.²⁷ Notably, those different strategies can also be translated and modified to fit to other types of VOC sensing devices (e.g., FET).

One interesting advantage of the hybrid systems is the ability to combine the properties of single material components into one system, and sometimes even generate new properties. For example, the combination of ionic liquids (ILs) with MOFs obtained a hybrid material with ionic conductivity and tuned sensitivity to VOCs (Figure 11a).³³⁸ This hybrid system combined the good chemical sensing capabilities of MOFs due to their intrinsic porosity with the ionic conductivity of ionic liquids. Upon adsorption of VOCs, the mobility of the IL's ion

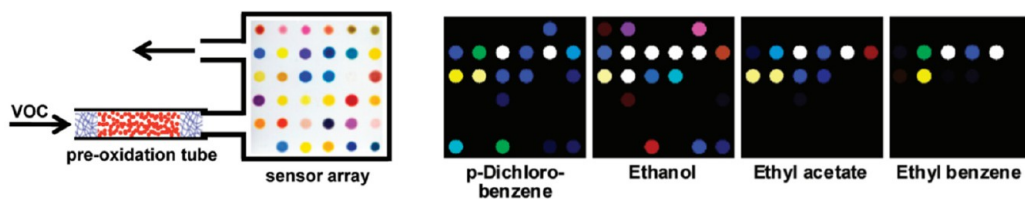


Figure 12. Schematic illustration of the preoxidation technique. A Teflon tube is packed with chromic acid to pretreat the VOC-containing gas sample before it passes over the colorimetric sensor array (left). Color difference maps for four indoor pollutant VOCs at their immediately dangerous to life or health concentrations (right). Reprinted with permission from ref 346. Copyright 2011 ACS.

pairs confined within the porous space of the MOF is disturbed, producing a measurable electrical response. The sensing performance of this hybrid was demonstrated in printed thin film capacitors under exposure to water, acetone and ethanol. Multiple parameters can be controlled to tune the chemical sensitivity and to boost selectivity to different VOCs including the MOF concentration and the chemical structure of MOF/IL system.

Hybridization has also proved useful for colorimetric and fluorescent sensing materials. One way of hybridization was achieved by immobilizing dyes into different matrices. The matrix holding the fluorometric and colorimetric dyes can contribute to improving the chemical selectivity, either by modifying the local chemical environment of the dye or by immobilizing the dye molecules in a sterically confining surrounding.⁴¹ This was demonstrated by the development of porous pigments in which chemically responsive dyes have been immobilized in a matrix of organically modified siloxanes.¹⁷² Jiang et al.³³⁹ reported an interesting hybrid system comprising PDA embedded within aerogel pores. The PDA–aerogel powder underwent dramatic color changes in the presence of VOCs (Figure 11b), facilitated through infiltration of the gas molecules into the highly porous aerogel matrix and their interactions with the aerogel-embedded PDA units. The PDA–aerogel composite exhibited rapid color/fluorescence responses and enhanced signals upon exposure to low VOC concentrations. Encapsulation of PDA derivatives with different headgroups within the aerogel produced distinct VOC-dependent color transformations, forming a PDA–aerogel-based sensing array. Another notable hybridization approach for the design of colorimetric sensing materials is based on the introduction of dyes into the voids or onto the surfaces of MOFs. Such chromophore/MOF hybrid structures show improved sensitivity to VOCs as well as stability due to the protection provided by the framework.^{340,326}

5. MAIN LIMITATIONS IN THE FIELD

5.1. Ultralow Concentrations and Low Reactivity of Targeted Analytes. Ideal VOC sensors have to perform as well as human/animal noses, which are acutely sensitive, starting at a 0.01 nM odor concentration, and highly selective, distinguishing at least one trillion different odors.³⁴¹ Mimicry of these natural systems, perhaps ambitious, could be used to address the difficulties of VOC sensing in real-world samples. Currently, most reported VOC sensors can detect ppm or sub-ppm level gases in the lab. However, some applications, e.g., diagnosis from exhaled breath and detection of explosives, require more sensitive sensors with LODs of ppb and even lower. Additionally, the chemical reactivity of VOCs in real life samples extends over a very wide spectrum; some are reactive (e.g., aliphatic amines) and therefore easier to detect, while others have low reactivity (e.g., nonpolar VOCs), making their

detection very challenging. Therefore, design strategies to enhance sensor sensitivity are sought. As already mentioned, one direction would be enabled by harnessing the nanostructuring approach,^{9,10,27} which provides superior surface areas and tuned chemical and physical properties. Additional methods include the commonly used high temperatures (mainly with metal oxides) and UV irradiation.³⁴²

To address the low VOC concentration challenge, preconcentration, preseparation, and target VOC marking methods can be considered.³⁴³ Microfluidic systems and microgas chromatographs (μ GC) could be the most promising methods for preseparation and preconcentration of gas mixtures (see section 6.2). Enrichment on solid adsorbents and membrane extraction could be a promising solution as well.³⁴⁴ In breath analysis, for example, a well-known porous polymer, Tenax TA, based on poly(2,6-diphenyl-*p*-phenylene oxide), is widely used as an adsorbent for preconcentration. As an example, Tenax was used for direct extraction and preconcentration of small molecule metabolites from a cell culture sample, such as *in vitro* extraction of metabolites obtained from cancer cells.³⁴⁵ This approach was shown to enable characterization of low-volatility cancer cell metabolites that could not be achieved otherwise.

Since the low reactivity of most VOCs (e.g., nonpolar compounds) makes their detection challenging, strategies to increase VOC reactivity before sampling could prove interesting. Colorimetric sensor arrays work well with reactive VOCs but do not show strong responses to less reactive ones. For example, common indoor air pollutants (e.g., alcohols, aromatic hydrocarbons, chlorocarbons, and some organic solvents) are generally not highly reactive and are not detectable at low vapor concentrations. Suslick et al.³⁴⁶ introduced an interesting method for improving the sensitivity of colorimetric sensor arrays for detecting and identifying less reactive VOCs by treating the analyte streams with strong acids (e.g., chromic acid loaded on silica) before they reach the sensing array (Figure 12). Preoxidation converts VOCs to chemically reactive species such as aldehydes, ketones, carboxylic acids, and quinones, which interact more strongly with the relevant colorimetric sensor elements. This results in an average ~ 300 -fold enhancement of sensitivity and a concomitant increase in the discriminatory power of the array.

5.2. Interfering Signals, Complexity of Samples, and Sensor Cross-Reactivity. One of the most intricate challenges in the field of VOC sensing (and sensing in general) can be briefly described as the (in)ability to detect the targeted signals precisely and reliably in complex environments. Most VOC sensors, especially modern soft ones, fail to meet the requirements for drastic (real-world) working conditions owing to their cross-reactivity (i.e., similar responses to different signals) and instability. In addition, the huge complexity of common real-world samples, which

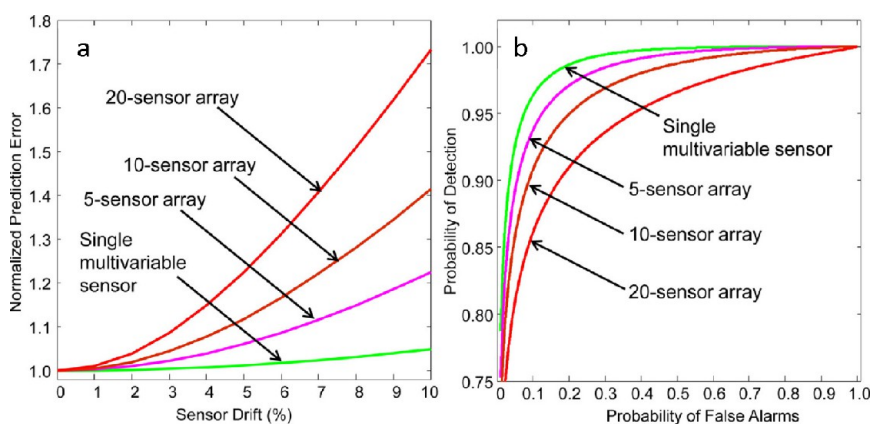


Figure 13. Simulation of long-term performance of individual multivariable sensors vs sensor arrays. (a) Prediction error and (b) probability of false alarms for an individual multivariable sensor with five outputs and for arrays of 5, 10, and 20 single-output sensors. Reprinted with permission from ref 354. Copyright 2016 ACS.

comprise very complex and dynamic mixtures of chemical and physical signals, adds to the challenge. Most of these signals are background and/or interference; some have drastic effects on the sensing system while others are neutral. Interfering signals can take different forms, starting from VOCs similar to the target and common inorganic gases, to humidity, temperature, and even strain/pressure in particular applications such as wearable devices. These signals must be carefully identified and neutralized for optimal sensing performance.

Haick and co-workers offered an interesting approach to overcome the effect of humidity interference in CNT-based sensors.³⁴⁷ Seven polycyclic aromatic hydrocarbons (PAH) with different aromatic coronae and side groups were used to coat the CNT sensing films. Using appropriate combinations of PAH/SWNT sensors, they showed that high sensitivity and accuracy can be obtained in discriminating polar from nonpolar VOCs in samples with widely different humidity levels (i.e., 5–80% RH). Another interesting way to neutralize humidity involved integrating a hydrophobic layer on the surface of doped PANI, resulting in a flexible sensor that could resist water at concentrations up to 350 ppm.³⁴⁸ In another study, MOFs were used to eliminate the effects of humidity on metal oxide-based VOC sensors, improving the detection of acetone under humidity interference.³⁴⁹ Many methods have been tried, but it is still unclear whether any are reliable outside the lab.

To address the interference caused by background gases and VOCs, pre-separation systems based on microfluidic and μ GC systems are expected to provide the best solutions (section 6.2). Temperature interference is more challenging since almost all sensing platforms are sensitive to temperature. Therefore, neutralizing temperature effects will probably come from advanced calibration methods, not from material and sensor design.

Overall, despite the numerous solutions provided for neutralizing interfering signals, this limitation still impairs progress in this field, especially the transition of devices from the lab to the real-world environment. We believe the solution will come from two different fronts. First, continuing the development of design concepts of sensing materials and devices targeting sophisticated components, that provide the ultimate interplay between sensing cross-reactivity, selectivity, and stability. The second would come from combining such sensing concepts with advanced ML strategies to provide the

most reliable and accurate calibration and sensing performance.

5.3. Instability of Sensors. Our nose can operate continuously and repeatedly for more than 70 years in various environmental conditions of temperature and humidity. Obviously, artificial systems linger behind the natural ones. The stability of the basic sensing materials and devices is crucial in designing reliable VOC sensing technologies. Multiple parameters such as relative humidity or temperature, and operational conditions such as applied voltage, can change the sensor's properties; this is manifested as drift.³⁵⁰ The instability of VOC sensors can be caused by changes in the intrinsic or extrinsic properties of the device such as structure alteration, phase conversion, poisoning, degradation, photo-bleaching, and bulk diffusion. These can eventually cause shifts in the baseline signal of the sensor over time. Moreover, continuous exposure to any VOC can instigate fouling, chemical alteration, or hysteresis, and (irreversible) nonspecific adsorption on the sensor surface.³⁵¹ For example, carbon black polymer composites lose stability because of swelling–deswelling-induced aggregation of the conductive filler, changing the electrical properties of the device. This stability problem is especially critical in long-term use. Careful design of sensing materials and determination of the sensor's optimal operating conditions should be carefully considered to minimize instability in each single sensor.

Even though there are multiple examples of drift compensation in gas sensor arrays,^{352,353} it is still a serious unsolved problem for long-term use because each sensor in the array has its own sensing material with its own drift, uncorrelated with other sensors.^{354,355} An interesting solution is suggested by single multivariable sensors (i.e., sensors with multiple output variables, also termed virtual multisensors),³⁵⁵ where instabilities in each output are correlated because only one sensing material and one transducer are used. These instabilities can be corrected more effectively than the uncorrelated drift of each sensor in an array. The advantage of multivariable sensors was demonstrated by comparing the predicted error in analyte concentration by a multivariable sensor with five outputs to arrays of 5, 10, and 20 single-output sensors. The five outputs of the multivariable sensor were calculated with a correlated drift in the range 0–10%. The same range of drift was used to calculate the prediction error of an analyte concentration for sensor arrays, but taking their

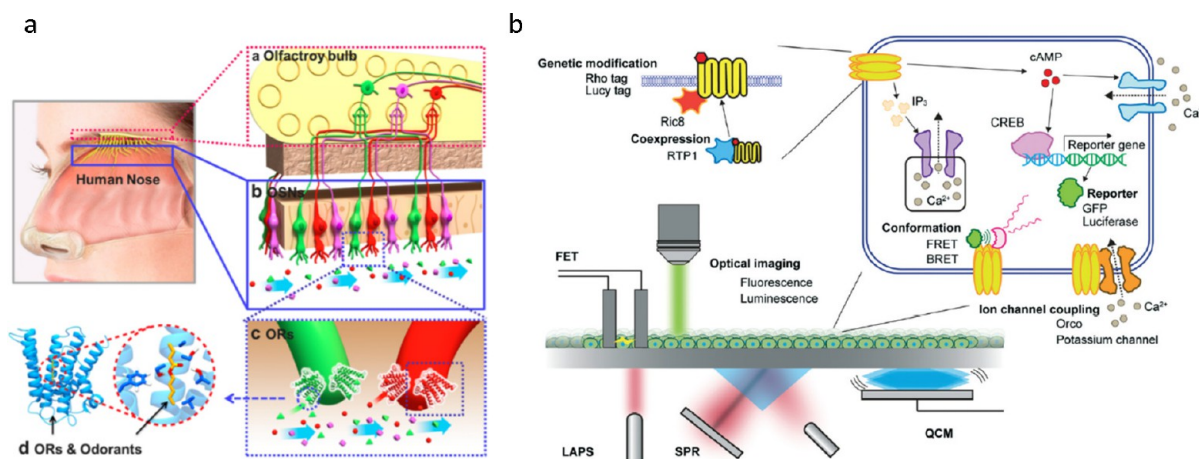


Figure 14. Biohybrid VOCs sensors. (a) Illustration of the human nose and olfactory bulb that include multiple olfactory receptors for odorant recognition. Reprinted with permission from ref 374. Copyright 2015 ACS. (b) Technologies and readout devices for sensitive odorant detection by cell-based sensors. The signal from the olfactory receptor can be amplified by connecting to an efficient pathway that transduces odorant binding to mediators such as cAMP and IP₃. LAPS light addressable potentiometric sensor. Reprinted with permission from ref 361. Copyright 2021 Royal Society of Chemistry.

uncorrelated drift into account (Figure 13). When the drift was zero (i.e., fresh sensors), the prediction errors for the multivariable sensor and the sensor arrays were the same, and minimal. However, with uncontrolled long-term drift, the sensor arrays lost their prediction accuracy much faster than the single multivariable sensor. These results show the potential of multivariable sensors to eliminate instabilities resulting from drift and noise.

5.4. Irreversibility. Irreversibility can be considered a form of instability. The big question would be is the irreversibility of sensors good or bad? In some types of sensors, named “dosimetric”, irreversibility of the sensor signal is desirable.⁴¹ These sensors are particularly required for ultratrace detection (ppb level and below) where cumulative or dosimetric sensing performance is necessary. One interesting example was developed using the chemical sintering of silver nanoparticles for VOC and gas sensing.^{174,356,357} Chemically induced nanoparticle sintering leads to changes in LSPR, and consequently in color, upon exposure to reactive VOCs. The cumulative nanoparticle sensors generally show sub-ppb detection limits for air pollutants such as HCOOH, HCHO, and other inorganic gases (SO₂ and NO₂) in 1 h exposures.

Dosimetric sensors are promising, but their use is limited to disposable arrays. With this in mind, irreversibility can cause serious problems in VOC sensing devices meant for continuous recording. In essence, it is a form of instability that causes changes in the chemical and physical properties of the sensing device. Such irreversible changes are difficult to compensate and can lead to inaccurate measurements after the first few exposure cycles.⁴⁰

The recovery time of sensors is strongly associated with irreversibility. In some cases, the sensor could be reversible, but only slowly. This would be problematic in highly dynamic applications that require high frequency sampling and monitoring of VOCs. Generally, the reversibility and recovery times of sensors can be improved by nanometric design and hybridization strategies. Other methods such as heating are commonly used to solve this problem.

5.5. Other Limitations. Other limitations in the development, operation, and application of VOC sensors are specific to certain types of VOC sensing materials and devices. Such

limitations include response time, unfavorable operating conditions (e.g., high temperatures), reproducibility, miniaturization, power consumption, and cost. Metal oxide-based sensors usually require high operating temperatures, limiting their use in some areas (e.g., flexible and wearable sensors that have temperature constraints). Many optical systems suffer from high power consumption and cost owing to the need for optical light sources, adequate optical path length, and a suitable optical detector.¹⁰⁰ Moreover, these requirements render such systems less amenable to miniaturization. Poor reproducibility is another limitation that can result from small variations in material composition and fabrication conditions. Careful calibration is required to address this.³⁵² Cost considerations are relevant during every step in the sensor's life cycle (i.e., design, fabrication, operation, and application). For example, printing is an excellent way to decrease fabrication costs and allow for easy high throughput production, but it is not compatible with all sensing materials.⁵⁶

6. ADVANCED AND EMERGING TRENDS

6.1. Biohybrid VOC Sensors. Natural olfactory systems in different species rely on a repertoire of receptors and combinatorial neural coding mechanisms to interpret the vastly complex chemosensory environment.³⁵⁸ Insects are reported to have 10–352 olfactory receptors,³⁵⁹ while mammals possess approximately 300–1300.³⁶⁰ By harnessing the VOC recognition elements found in nature, biohybrid sensors have been introduced.³⁶¹ chemical sensors that directly integrate biological molecules and cells for ultrasensitive detection of VOCs. These sensors use olfactory receptors from organisms and can be classified as cell-based or protein-based (Figure 14).³⁶¹ Cell-based sensors use cells that express olfactory receptors on their membranes, whereas protein-based sensors directly integrate olfactory receptors into a device. In both systems, odorant receptors play a central role as they recognize the molecular configuration of odorants and trigger a downstream signal that is recognized by other sensory components.^{362,363} In cell-based sensors, the olfactory signals are transduced and amplified through two signaling pathways, the cAMP and the IP₃ routes.^{341,364–369} The sensitivity of

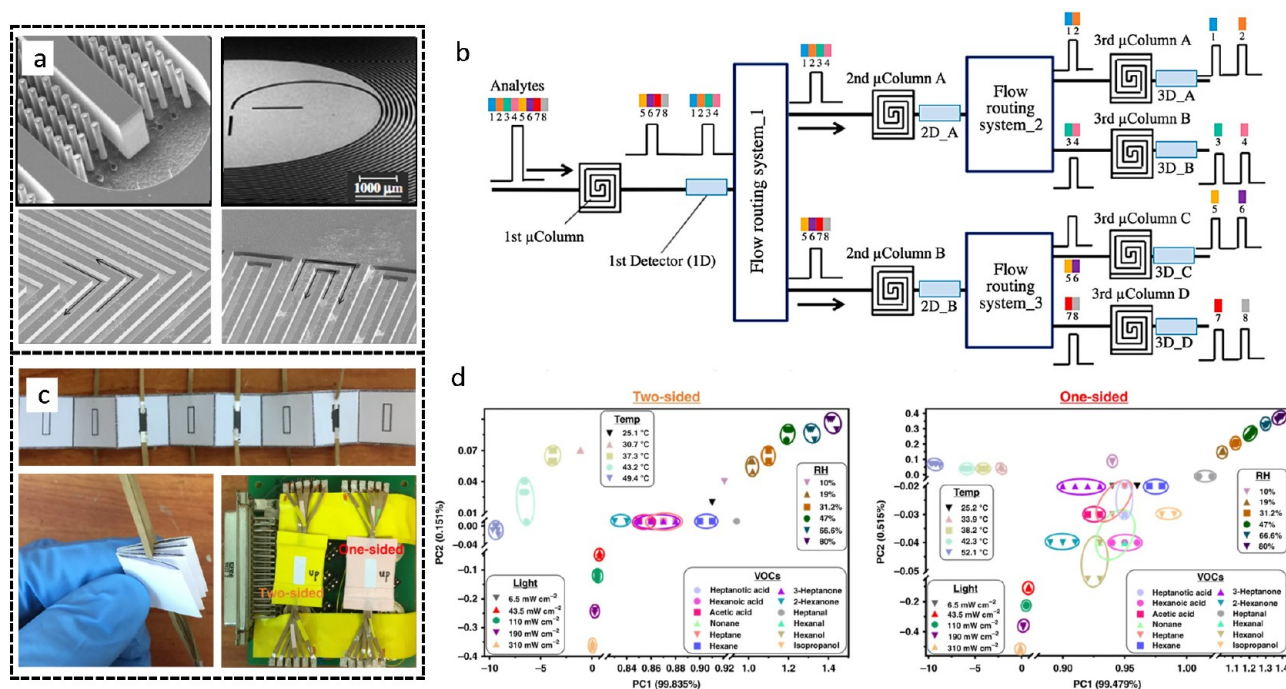


Figure 15. (a) SEM images of some typical layouts of μ -columns investigated in the literature: semipacked column with embedded microposts, circular spiral, square spiral, and radiator. Reprinted with permission from ref 388. Copyright 2013 Elsevier. Reprinted with permission from ref 385. Copyright 2009 IEEJ. Reprinted with permission from ref 389. Copyright 2006 Elsevier. (b) Schematic representation of an automated 3D- μ GC system with three different levels of separation. Reprinted with permission under a Creative Commons Attribution CC BY License from ref 343. Copyright 2020 MDPI. (c) Photographs of origami-based sensors made from paper and graphene-based ink. (d) Two-dimensional canonical score plots of the responses of two-sided and one-sided origami sensors to multiple stimuli. Reprinted with permission under a Creative Commons Attribution CC BY License from ref 387. Copyright 2019 Springer Nature.

biohybrid sensors can be improved by careful selection of the transduction principles and the methods by which the olfactory systems are immobilized on readout devices.³⁶¹

The performance of biohybrid odorant sensors could potentially exceed that of the human nose, especially in their ability to quantify odors. Another advantage of biohybrid sensors is the degree of freedom in choosing odorant receptors on the sensing elements. Since the set of odorant receptors has been optimized throughout evolution, the receptors in animals of different species have become specialized for specific odorants. For example, there are odorants that dogs or mice sense but human beings cannot,³⁷⁰ and vice versa. Therefore, biohybrid sensors could be engineered to combine odorant receptors from various species in a single device, allowing us to develop an olfactory sensing system superior to that of any existing animal.

Bioengineering has a key role in the development such sensors. In one example, an odorant receptor gene was genetically modified to fuse with green fluorescent protein (GFP) and luciferase genes. The resulting fusion protein underwent conformational changes upon binding to odorants, allowing energy to be transferred from luciferase to GFP, resulting in fluorescence emission, directly detected by the system.³⁷¹ An alternative approach to detecting the binding of odor molecules is to fuse olfactory receptors with ion channels.³⁷² Here, odorant binding opens the ion channel and causes ion influx into the cells, which is detected using ion-responsive fluorophores. More importantly, bioengineering approaches allow the sensitivities of biohybrid sensors to be improved. One major approach has focused on increasing the

sensor element density (i.e., the density of functional olfactory receptors on the cell surface).

This area is promising, but many problems remain to be solved before such technologies can be commercialized. It is challenging to produce olfactory receptors on a large scale. Limited receptor stability, receptor saturation, and lack of receptor regeneration strategies hamper applications that require continuous real-time measurements. The structure and function of olfactory receptors depend on their native environment, which further limits strategies to aqueous-based systems and reduces the number of stable display formats available for olfactory receptors.³⁷² There is also a need to focus on the stability and lifetime of biohybrid-based sensors. For instance, olfactory receptor stability can be improved by incorporation into microfluidic devices that keep the receptor in a controlled fluidic environment.³⁷³

6.2. Microanalytical Systems Combined with VOC Sensors. As mentioned earlier, there are several strategies to improve the selectivity of single-based gas sensors such as specially functionalized surfaces, doping of nanomaterials, multicomposite (or mixed) materials, or temperature cycling.³⁴³ However, these strategies are suited to very specific types of sensing materials and devices, which compromises their modularity and flexibility of implementation. In this context, a combination of nanomaterial-based sensors and conventional VOC separation methods such as GC³⁷⁵ could provide a universal platform for selective VOC sensing.^{343,376–381} Here, μ GC and microfluidic systems become very advantageous.

6.2.1. Microgas Chromatographs. Advances in micro-electromechanical systems (MEMSs) have allowed conven-

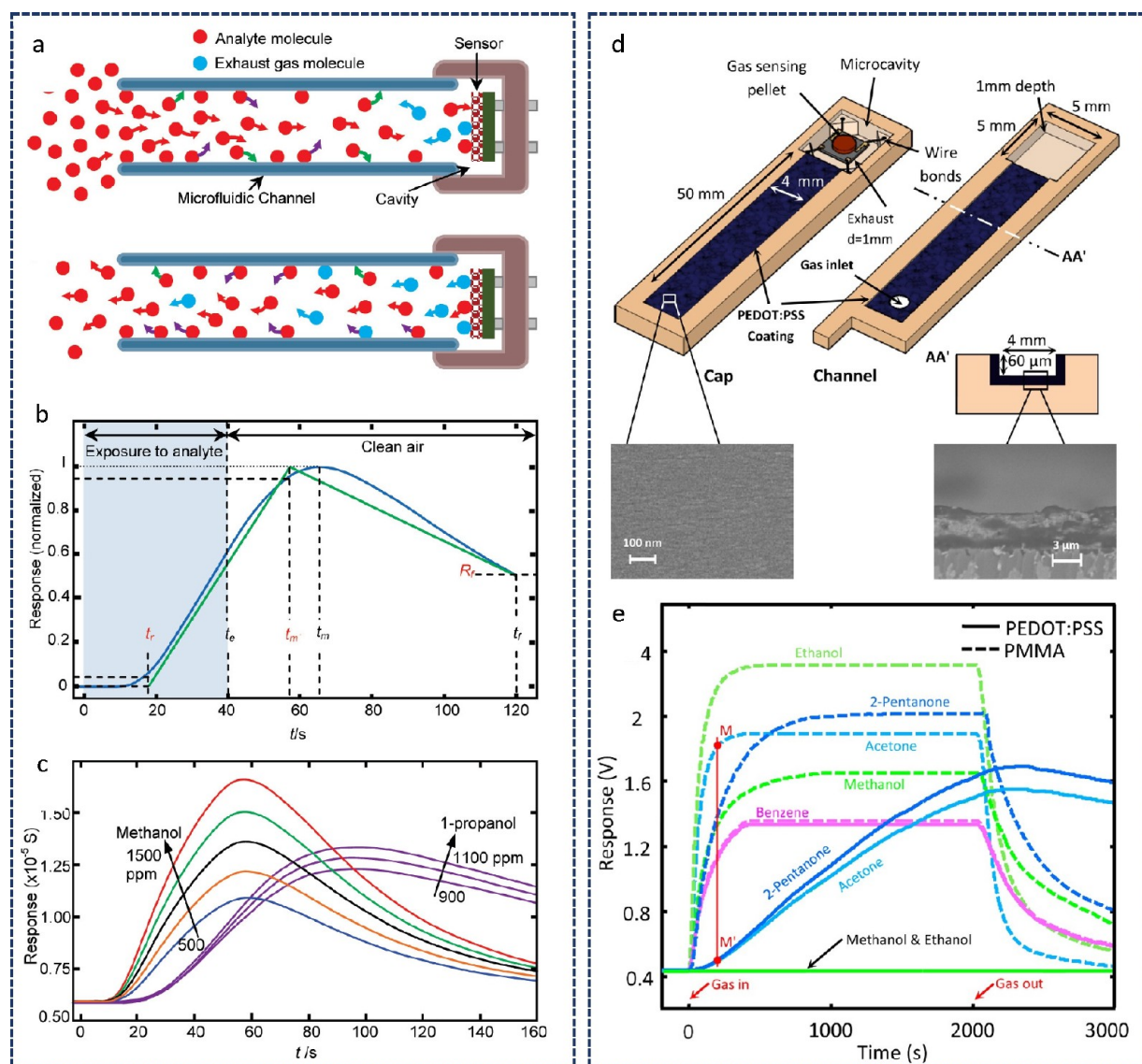


Figure 16. (a) Diffusion of analytes into a microfluidic channel. Physisorption to the walls reduces the effective diffusion rate. (b) Experimental normalized response profile and its “line diagram”. (c) Response profiles obtained for methanol and propanol at the stated analyte concentrations in the gas chamber. Reprinted with permission from ref 380. Copyright 2010 ACS. (d) Schematic presentation of the PEDOT:PSS-coated microfluidic channel integrated with a gas sensor. (e) Temporal sensor responses recorded for the stated VOCs using control (PMMA) and coated (PEDOT:PSS) channels with 40 s exposure times. Reprinted with permission under a Creative Commons Attribution CC BY License from ref 396. Copyright 2017 Springer Nature.

tional chromatographic components to be miniaturized in compact and portable systems, leading to the introduction of microgas chromatographs (μ GC) that can be used for on-site and real-time separation of VOCs.^{382,383} Such sensing systems usually include microfabricated components for injection (μ -injectors), separation (μ -columns), and detection (μ -sensors). The most critical component in μ GC is the microfabricated column, which must be carefully designed to optimize the separation of analytes. Like conventional columns, μ -columns are strategically coated with a stationary phase to optimize the separation of vapor analytes,³⁸⁴ but they are several orders of magnitude shorter, noncylindrical, and normally microfabricated on top of planar substrates using a chip-based configuration.³⁸¹ The separation efficiency of μ -columns depends on several factors including column geometry (e.g., cross-sectional area, shape, topology), stationary and mobile phases, operating temperatures, and flow rate.³⁴³ Metal, glass

polymer, and silicon-based materials are the most common substrates for μ -columns because they have suitable physical, thermal, and chemical properties.³⁸⁵ For the stationary phase, polymers such as polydimethylsiloxane (PDMS) are generally preferred owing to their ease of handling, chemical inertness, and high porosity, which contribute to efficient separation, especially of nonpolar analytes.³⁸⁶ Figure 15a,b presents typical layouts of μ -columns investigated in the literature, and a schematic representation of a 3D- μ GC, which involves multiple columns and detectors and offers better separation than simple systems.

Haick and co-workers³⁸⁷ reported an approach that can be categorized under this type of microanalytical system. A chromatography-like system was developed using origami structures fabricated from paper (Figure 15c). The authors used multiple layers of paper to obtain a hierarchical structure that manipulated VOC transport toward the printed ink-based

sensing layers. The gas was delivered to the sensing area by passing through the multiple layers of paper that constituted a microporous medium for separation. This design provided time-space-resolved responses and generated highly discriminative pattern recognition features. It enabled the simultaneous detection of different VOCs as well as their identification in complex mixtures (Figure 15d). Of special importance, this device proved highly effective in differentiating among structural isomers and chiral enantiomers of VOCs at 10 ppm.

6.2.2. Microfluidic Systems. “Microfluidics” denotes the study of fluids at the microscale using channels with dimensions of tens to hundreds of micrometers fabricated on small-scale chips.³⁹⁰ In VOC analysis, microfluidics has been used for sample manipulation prior to examination by analytical techniques and sensors.^{343,376} This is considered a promising alternative to μ GC for selective identification of VOCs in complicated gas mixtures.^{391–393} It offers numerous advantages such as requiring minimal quantities of samples for testing, precise control over experimental conditions, integration of multiplexed processes, short analysis time, and suitability for in situ operation.³⁷⁶ Microfluidics makes VOC sensors selective by separating analytes by their rates of diffusion through a specially coated microfluidic channel (Figure 16).³⁹³ Differences in molecular diffusion and surface physisorption of analytes control their transport toward the sensing area.³⁹⁴ Microfluidic channels are normally shorter than chromatographic columns, do not require a carrier gas, and can operate at room temperature.³⁹⁵ These systems can readily be integrated into miniaturized devices using widely available and low-cost microfabrication processes such as photolithography, soft-lithography, and micromilling.^{376,390} This allows for the design of customized microchannel geometries and for the integration of different steps in sample manipulation in compact devices.

The selectivity and performance of microfluidic devices rely on several factors. As in chromatographic columns, the geometrical properties of the channel are designed to optimize selectivity while ensuring acceptable response and recovery times.³⁸⁰ Channel composition and coating material are also key to the separation capabilities of microfluidic devices.^{392,396,397} PDMS has been the most widely used component in preparing microfluidic systems because it is affordable, transparent, and nontoxic.³⁷⁶ For coatings, chemically inert and highly porous materials that adsorb analytes efficiently are commonly used (e.g., poly(3,4-ethylenedioxythiophene):poly(styrene-sulfonate) (PEDOT:PSS) and Parylene C).³⁴³ Several methods have been proposed to change the roughness of the channel and improve analyte separation. Notably, nanotechnology and nanomaterials provide interesting opportunities for better analyte separation. For example, MIP nanoparticles designed for acetone recognition were integrated into a 3D-printed channel platform to obtain high selectivity for acetone against various VOCs (alcohols, ketones, nitrile, and aromatic compounds).³⁹²

To use the segregation power of microfluidic-based systems, the transient response of the sensor is normally assessed over multiple analytes.³⁴³ The analysis of results is based on the principle that different analytes lead to small shifts in the response signal while different analyte concentrations change its amplitude. Generally, two or three features are extracted from the response pattern to represent different analytes as separate clusters in a 3D or 2D feature space. Like e-noses,

microfluidic devices are normally trained to identify and classify VOCs of interest. Results obtained from unknown samples can then be related to existing clusters of certain analytes.³⁹⁴

Nanofluids have been less involved in gas sensing although they have found multiple applications in general chemical analysis.³⁹⁸ It is anticipated that this technology will provide more opportunities for VOC sensing, and, therefore, it is recommended to seek for combinations of miniaturized VOC sensors and nanofluidic systems.

6.3. Artificial Intelligence and Machine Learning for Advancing VOC Sensing. Humans develop their perception of odors by experience, allowing them to recognize different chemicals. The same applies to artificial VOC sensing and smelling systems. For selective sensing of specific VOCs in small samples, direct measurement with minimal signal processing and data analysis can be used owing to the simplicity of the target analyte. However, advanced algorithms and analytical methods are needed to meet the requirement for sensing complex combinations of VOCs, especially under interfering conditions. These algorithms train artificial systems to understand their environment and to optimize their decisions. ML³⁴⁶ in particular, can be used to neutralize or minimize the effects of irrelevant data while extracting useful information on the VOCs of interest.³⁴⁶ This allows for better selective detection of desirable VOCs and minimization of interfering effects and background noise.

ML can be divided into two categories: unsupervised and supervised. Unsupervised learning (i.e., without a pre-existing library) is intended to uncover hidden patterns in unlabeled data.³⁹⁹ It shows clustering of data but does not provide predictive classification. This approach has been extensively involved in the analysis of VOC sensing data, using different methods including principal component analysis (PCA) and hierarchical cluster analysis (HCA). Supervised learning usually involves classification methods that typically employ predetermined data (a training set or library) to classify unknown samples using linear or other classifiers. Common ways of maximizing discrimination include hypothetical methods (e.g., linear discriminant analysis (LDA)) or empirical predictive methods (e.g., support vector machines (SVMs)). Linear models can comprehend linear relationships between features and labels, but are not suitable for studying the nonlinear relationships that occur in biological samples. Nonlinear models are often more accurate than linear ones. These techniques can be grouped into three subcategories: kernel-based algorithms, tree-based algorithms and neural networks.⁴⁰⁰ Good summaries of the most useful algorithms in VOC sensing can be found elsewhere.^{37,400} Many VOC sensing areas have benefited from such ML algorithms including environmental monitoring, disease diagnosis,^{9,12,187} food and agriculture,^{401,402} and explosive detection.⁴⁰³

A prominent example of the use of ML methods for VOC sensing is in the healthcare applications, especially in disease detection. Patients usually seek healthcare help during routine follow-ups or when symptoms appear related to a health status change. However, in both cases the timing, which has a significant influence on the success of treatment and the recovery rate, is not optimal. A combination of VOC sensors with ML methods could help mitigate this difficulty through body monitoring and early detection of diseases.⁴⁰⁴ In particular, artificial neural networks, which are based on deriving a set of variables (hidden layers) as a linear

combination of the input layer and building a nonlinear model from those layers, have been extensively used for breath analysis and have successfully detected multiple diseases such as lung cancer,⁴⁰⁵ breast cancer,⁴⁰⁶ and multiple sclerosis.⁴⁰⁷

6.4. Highly Informative VOC Sensors. Conventional sensing platforms primarily comprise one or two sensing modules capable of performing one function or sensing one parameter. This low functionality is a major obstacle facing general sensing applications and VOC sensing in particular. Given the diversity of chemical and physical signals, low-functional sensing platforms cannot satisfy the increasing demand for complicated sensing capabilities in real-world applications. Even with the help of ML methods, advanced sensing systems are still required. These considerations have led to the definition of “highly informative sensors”, viz. sensors that provide extended amounts of chemical and physical information about their environment. The extended information, obtained by combining multiple sensory outputs, can serve as a basis for superaccurate detection of target signals. This provides many advantages such as quantification of individual components in gas mixtures, rejection of interference, and self-correction against environmental instabilities. However, reports concerning highly informative sensors are still not prevalent in the literature. Authors refer to this feature/property using different “close” concepts including multivariable/multiparametric, dual-modal and multimodal, and multifunctional. However, each of these terms denotes just one type of highly informative device, so they should not be used interchangeably.

Multifunctional devices are an important type of highly informative sensor. This term has mainly been used to denote sensors capable of sensing different signals (different VOC groups and/or even combination of VOCs, pressure, temperatures, and humidity). In a recent study, Wu et al.⁴⁰⁸ reported a PANI-based multifunctional flexible sensor that could detect and distinguish among different stimuli including pressure, temperature, and VOCs. Interestingly, all the sensing properties explored were stable under different bending/strain states. In different demonstrations, the multifunctionality was even extended to include additional “nonsensing” functionalities such as self-healing (though this could be considered self-sensing),^{217,409,410} a property akin to the miraculous survival capability of all living organisms that allows them to recover from structural and functional injuries incurred during their lifetimes. Like its benefits in other sensing devices,^{411–418} this property could be advantageous for improving the stability and reliability of modern VOC sensors.

A very promising approach for highly informative sensors relies on multivariable sensing materials.³⁵⁴ These materials exhibit multiresponse mechanisms to different VOCs, coupled with a multivariable transducer to produce independent outputs corresponding to the different sensing responses. In essence, the term “multivariable” is equivalent to multiparametric, describing sensors that provide multiple independent sensing features/variables. Haick and coworkers have discussed the importance of this property in FET sensors that can identify VOCs by changes in their characteristic features/parameters (e.g., mobility, threshold voltage, on-current).^{89,90,186,187,289} Many methods have been used for generating a multivariable response, or to extract multiple sensing parameters from a single device. For example, impedance spectroscopy (instead of a simple resistance/capacitance single measurement) and temperatures cycling

(instead of measuring at one temperature) can provide additional independent sensing variables.³⁵⁵ Multivariable device will expectedly grow and complement the conventional single-output devices and sensor array strategies (Figure 17).

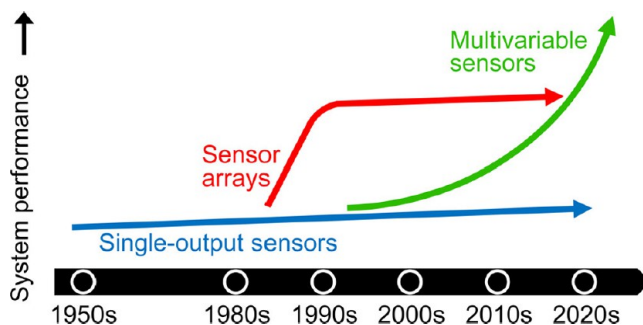


Figure 17. Multivariable sensor technologies to complement and ultimately to replace conventional single-output sensors and sensor arrays. Reprinted with permission from ref 354. Copyright 2016 ACS.

If different transduction methods/modes are used to generate multiple sensing features, the device could be also called multimodal. In one interesting example, Zhang et al.⁴¹⁹ demonstrated a dual-mode gas sensor for simultaneous and independent acquisition of electrical and mechanical signals from the same sensing device. The sensor seamlessly integrated a graphene-based FET with a piezoelectric resonator. Dual signals from independent physical processes, e.g., mass attachment and charge transfer, were combined to recognize and quantify VOCs of interest. The dual mode performance led to an improved understanding of the interactions between graphene and various gas molecules. Besides its practical functions and its ability to provide more sensing information, this device can serve to investigate the mechanism involved in the sensing process quantitatively.

Highly informative sensors generate an extended amount of data about the targeted system. Fusion of data from multiple sensory components shall be considered an emerging strategy in various domains. For example, in healthcare, pulse rate, respiration rate, blood pressure, and body temperature, sometimes enhanced with SpO₂, glucose, or other signals, are displayed and can be combined into an “early warning score” to facilitate the detection of deteriorating health. These strategies are proving highly useful for improving the performances of sensing systems and obtaining more reliable evaluations. We expect to see more translations to the field of VOC sensing soon. The huge variety of combinatorial sensing opportunities can enhance the reliability and accuracy of VOC sensing. However, gas detection scenarios with complex and unknown gas environments and background signals could require extensive data collection, modeling, and ML.

7. REPRESENTATIVE APPLICATIONS OF VOC SENSORS

The exponential increase in efficient, simple, and cheap VOC sensing materials and devices has led to the introduction of advanced sensing technologies and to their integration in different fields. This section highlights a few promising topics including the artificial nose, gas sensors that serve the “internet of things” paradigm, and disease detection and body monitoring.

7.1. The Artificial Nose. Since its description in 1982,¹⁶⁶ the artificial nose, commonly called electronic nose, has been developed to digitize olfaction by replicating and mimicking natural abilities for detecting and decoding VOCs, gases, and aromas.¹⁶⁷ Natural olfactory systems are more discriminatory and sensitive than other sensory systems. For instance, the human nose can discriminate more than one trillion different olfactory stimuli.⁴²⁰ Since humans have only 400 intact olfactory receptors,⁴²¹ it is impossible to attribute this ability to the “lock and key” model, in which each receptor responds to only one odorant.⁴²² Accordingly, natural olfactory systems interpret VOCs and aromas through a relatively limited number of odorant receptors with different ligand binding affinities, followed by computational processes inside the brain. Similarly, the artificial nose is designed on the basis of cross-reactive chemical sensor arrays, incorporating a variety of sensing materials, combined with complex algorithms for generating distinct response patterns and classification of VOCs.¹⁶⁶ This is usually followed by a training or learning phase that involves targeting preclassified samples under different conditions, defined on the basis of the application.⁴²³

Artificial noses have adopted a variety of sensing arrays.⁴²² Nanoengineering and nanohybridization have proved successful in designing the wide variety of materials required for the array. Electrical arrays are mostly based on one conductive material/backbone (e.g., metal particles, carbon materials, conductive polymers) functionalized with different surface groups to tune sensitivity,¹⁶¹ although other combinations can be found such as arrays based on CNTs and GNPs.⁴²⁴ For optical/optoelectronic noses, the arrays are mostly based on different dyes with a range of chemical interaction capabilities (see Figure 9).^{37,41} Other optical noses rely on optical fibers employing different materials such as polymer coatings and functionalized microspheres.⁴²⁵ Gravimetric arrays have also been used as artificial noses, relying mainly on QCM or SAW sensors.⁵ Here, the required cross-reactivity is produced by a variety of coatings (sensitizers) loaded on to the surface of the resonator (metal oxides, polymers and MIPs, carbon materials, self-assembled monolayers, and other hybrids).⁵ These modifications have proved very useful for tuning selectivity and sensitivity toward different analytes and eventually generating cross-reactivity.

To date, many artificial noses have been commercialized and are being used for diverse tasks.^{4,426} One major area is the food and beverage industry, where the primary role of the artificial nose is to assess the quality of a product during manufacture, shipment or storage. Many of these tasks are traditionally performed by highly trained human sensory panels, which can be subjective and prone to fatigue.¹⁶⁶ As an alternative to these panels, artificial noses can help to quantify multiple food and drink related processes including microbial fermentation,⁴²⁷ fruit and vegetable ripening,⁷ manufacture of bread and yeast products,^{428,429} dairy products,⁴³⁰ and wine and alcoholic drinks.^{431–434}

Food spoilage can also be monitored by artificial noses.^{435,436} The simplest methods that we use daily to verify the edibility of food are based on sight and smell, thanks to the millions of years of evolution that have rendered human noses extremely sensitive to spoiled food.¹⁶⁶ However, the gold standard for identifying unhealthy bacterial contamination in food is to culture samples in a lab, which is both labor and time intensive. The time lag in monitoring means that most incidents of contamination are not detected until after

shipment.⁴³⁶ Artificial noses have been used to detect spoilage/freshness rapidly in eggs, meat, fruits, and vegetables.^{436–439}

There are numerous nonfood applications for artificial noses as well. For instance, the devices can be used for environmental monitoring (real-time evaluation of pollutant gases emitted from industrial plants).^{433,440} Devices near a source of pollutants can warn when the concentration rises above a certain threshold or when hydrocarbons are released into soil or wastewater streams.^{441,442} Artificial noses can be used also for targeting landfills,⁴⁴³ composting facilities,⁴⁴⁴ and wastewater treatment plants,⁴⁴⁵ and in the production and analysis of pharmaceuticals (e.g., antibiotics produced in cell cultures), drug screening, and other applications.^{299,446,447}

Despite considerable and sustained efforts, it remains challenging to detect and discriminate odor molecules reliably and precisely in complex gaseous samples where many odorant species coexist, especially under varying environmental conditions such as humidity and temperature. The limitations discussed in section 5 also remain to be addressed before superior artificial noses that can match the performance of biological olfactory systems are achieved.

7.2. VOC Sensors Serve the “Internet of Things”

Paradigm. The IoT provides a huge network, an ecosystem, of connected things (machines, objects, people, plants, etc.) that gather, store, and use data about their environments.⁴⁴⁸ Sensors constitute a major part of this connection, providing for continuous monitoring of the state of things through emitted signals.⁴⁴⁸ At present, physical sensors dominate IoT applications, but we expect to see more involvement of chemical VOC sensors soon.³⁵⁴ They will expectedly contribute to multiple sectors such as agriculture, industry, and environmental monitoring.³⁵⁴ Using such IOT-enabled sensors we can evaluate the state/condition of an object/machine and then use this information to generate ML-enabled feedback and decisions. For example, VOC sensors are envisioned as integral parts of wearable accessories or portable devices that continuously monitor the environment, or human body.¹⁶ Users or stakeholders will be part of an IoT network that provides fast and continuous access to sensing information.¹⁴ The appearance of any abnormal condition can be detected, reported and treated in the most time-efficient way. This will enable interactive and evolving decisions to be made and used for evidence-based analysis and recommendations, and for personalized monitoring.⁴⁰⁴

The top requirements for modern sensors for IoT and industrial Internet applications include the following: (i) reliability, to provide accurate readings under different background conditions; (ii) low power or ideally self-powered; (iii) low cost, to accommodate the need for large numbers; (iv) miniaturization - sensors will be optimally integrated in high density and “seamlessly” (i.e., not interrupting their targets); (v) appropriate real-time communication capability; and (vi) data security (especially when data are gathered from human bodies).³⁵⁴ The development of VOC sensors combining all these properties will be highly valuable for advancing the IOT-enabled VOC sensing technologies.

Agri-tech is a growing technology segment in which the IoT is expected to show strong involvement, with the aim of optimizing the global food and agriculture industry.⁷⁸ In particular, greenhouse cultivation, which has progressed from simple covered rows of open-fields crops to more sophisticated controlled-environment agricultural facilities,⁴⁴⁹ will benefit

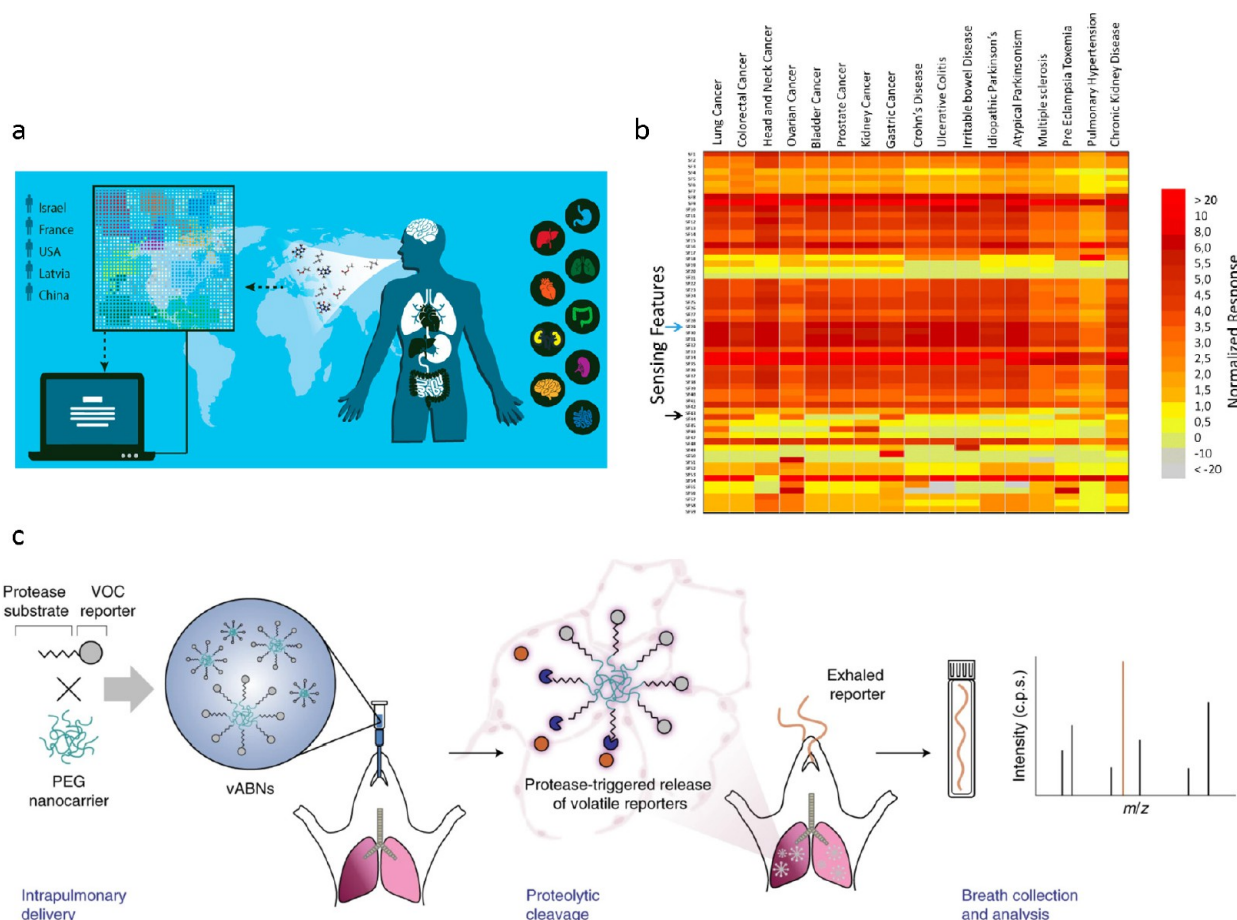


Figure 18. (a) Schematic representation of the concept and design of the study, which involved collection of breath samples from 1404 subjects in 14 departments in nine clinical centers in five countries (Israel, France, USA, Latvia, and China). The population studied included 591 healthy controls and 813 patients diagnosed with one of 17 diseases. Breath samples were analyzed with GC–MS to explore their chemical composition and with an artificially intelligent nanoarray for disease diagnosis and classification. (b) Heat map of sensing features, extracted from 20 different nanomaterial-based sensors, showing the mean responses for each of the 17 diseases. Reprinted with permission from ref 424. Copyright 2016 ACS. (c) Schematic picture showing the engineered nanoparticles that can be delivered into the lung for the release of detectable reporter VOC under specific respiratory disease conditions. Reprinted with permission from ref 475. Copyright 2020 Springer Nature.

greatly from IoT systems. It is well-known that plants produce a variety of metabolites including VOCs, which participate in a wide range of ecological functions as diverse as plant-to-plant communication, defense from predators, indirect defense against insects, pollinator attraction, thermo-tolerance, and environmental stress adaptation.⁴⁵⁰ Therefore, continuous monitoring of VOC emissions provides an excellent way of assessing the physiological state and health conditions of plants and the general performance of a greenhouse cultivation facility.

Many other examples of VOC sensors integrated with IoT systems can be found in the literature, targeting multiple applications such as atmospheric environment monitoring,⁴⁵¹ indoor air monitoring,⁴⁵² water pollution,⁴⁵³ materials classification,⁴⁵⁴ and improving energy efficiency in smart buildings.⁴⁵⁵ In the next section, we will discuss one important example, which is used to monitor the state of a human body through VOC emission, leading to promising opportunities for early disease detection and optimal treatments and interventions.

7.3. Disease Detection and Body Monitoring. Analysis of VOCs offers opportunities for human body monitoring and

detection of abnormal conditions such as cancers, kidney diseases, and neurodegenerative diseases.^{11,456} This is based on the fact that VOCs are products of metabolic and pathological processes at cellular and tissue levels and they are strongly affected by diseases and body conditions.^{15,456–460} These VOC-based methods are expected to be noninvasive and inexpensive in worldwide implementation (especially in developing countries) while providing: (i) screening of high-risk populations for emerging diseases; (ii) early detection and prediction of diseases; and (iii) evaluation and monitoring of treatment efficacy.^{10,11} However, the complexity of VOC samples emitted from the body presents a major challenge. It has been shown that healthy subjects secrete almost 2600 VOCs.^{17,459,461} This combination includes both endogenous and exogenous VOCs belonging to multiple families including hydrocarbons, alcohols, aldehydes, ketones, esters, nitriles, and aromatic compounds.^{17,459,462}

Several articles and reviews have addressed the relationship between VOCs and body conditions. VOC patterns and combinations characterizing the healthy body,⁴⁵⁹ chronic diseases,^{463,464} different types of cancer,^{197,465–468} infectious diseases,^{469–471} and mental disorders⁴⁷² have been identified.

A wide variety of technologies have been involved in such studies, from conventional GC–MS to advanced artificial noses (e.g., electronic nose^{201,407,424} and colorimetric nose^{473,474}). In a recent study, Haick and co-workers⁴²⁴ described a chemiresistive array of CNTs and GNPs for classifying 17 diseases in 1404 subjects via pattern analysis of exhaled molecules (Figure 18a,b). Despite the huge sample size, which was collected from different countries, clinical and environmental factors that could confound the performance of the sensor array did not affect the results significantly.

Despite significant developments, the field of precise clinical diagnostics still needs to overcome many bottlenecks and to address emerging conditions and diseases. Advanced approaches for improving the detection capabilities under interfering environments are highly required. For example, an interesting way to improve the detection capabilities of diseases is by using reporter VOCs. This was demonstrated for the detection of pulmonary diseases. Here, the authors engineered breath biomarkers for respiratory disease by local delivery of protease-sensing nanoparticles into the lungs.⁴⁷⁵ The nanoparticles released volatile reporters upon cleavage by neutrophil elastase, an inflammation-associated protease with elevated activity in lung diseases. After delivery into mouse models with acute lung inflammation, the volatile reporters are released and expelled in breath at levels detectable by mass spectrometry (Figure 18c). These breath signals can identify diseased mice with high sensitivity as early as 10 min after nanosensor administration.

8. SUMMARY AND FUTURE PERSPECTIVES

This review has broadly surveyed the field of VOC sensing. In particular, it has focused on the most recent promising directions ranging from the design of sensing materials, relying mainly on nanomaterials and related hybrid structures, to emerging and advanced sensing systems and applications. The promising solution-enabling strategies were highlighted and the main limitations of the field were carefully discussed. Despite the enormous progress made in recent decades and the countless demonstrations of sensing systems, VOC sensing technologies still need to satisfy the requirements for some envisioned applications such as disease detection through body fluids, wearable systems for body monitoring, and indoor detection and identification of toxic VOCs. Therefore, the next generation of VOC sensors should be focused on improving sensitivity and selectivity either by applying advanced materials and structures or by extracting target signals effectively by advanced ML algorithms.

More VOC sensors are expected to be involved in our daily environments, especially with the exponential growth in cheap, miniaturized, and printable sensing materials and devices. Optical VOC sensors will particularly benefit from rapid improvements in digital imaging and computational power, suggesting that future optoelectronic noses will take advantage of ordinary cell phones linked to Bluetooth readers holding sensor arrays. Altogether, we expect soon to see more sensing materials and devices with superior performance and to witness the emergence of additional VOC sensing applications. For example, VOC sensors applied in wearable devices⁴⁷⁶ have the potential to provide types of electronic skin that enable the sense of smelling in robotics and prosthetics. For such applications, VOC sensing systems must become softer, flexible, and even stretchable. Extensive research is needed to

push such emerging VOC sensing applications toward the market.

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